

Optimising Selberg's method for critical zeros

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Abstract

We revisit Zhuravlev's 1974 quantitative form of Selberg's sign-change method for the zeros of the Riemann zeta function on the critical line. Zhuravlev's work, originally in Russian and little known in the West, established an explicit positive proportion of such zeros. Using modern techniques we optimise the arithmetic mean value at the heart of the method, which has an exact closed form for reciprocal-square-root coefficients and is improved further by a positive-semidefinite family of mollifiers. We thereby obtain the strongest result yet known from Selberg's method, proving that at least 7% of the non-trivial zeros of the Riemann zeta function lie on the critical line.

1 Introduction

Let $N(T)$ denote the number of non-trivial zeros $\rho = \beta + i\gamma$ of the Riemann zeta function $\zeta(s)$ with $0 < \gamma \leq T$, counted with multiplicity, and let $N_0(T)$ count those with $\beta = \frac{1}{2}$. We write

$$\kappa_0 = \liminf_{T \rightarrow \infty} \frac{N_0(T)}{N(T)}.$$

Hardy proved that infinitely many zeros of the zeta function lie on the critical line, and Hardy–Littlewood obtained $N_0(T) \gg T$ [9, 10]. Selberg's theorem $N_0(T) \gg T \log T$ was the first positive-proportion result [18, 19]: his method detects sign changes of Hardy's real function after multiplication by a non-negative mollifier. Levinson's argument-principle method and Atkinson's two-point sign test are analytically different; Section 2 compares the three approaches, as well as the recent pair correlation method, and places Zhuravlev's work within the Selberg tradition.

This paper has two aims. First, we give a corrected and self-contained account of Zhuravlev's 1974 paper [22], which is written in Russian and little known in the West. Second, we optimise the arithmetic at the heart of his and Selberg's method. Our main result is the following.

Theorem 1.1. *At least 7% of the non-trivial zeros of $\zeta(s)$ lie on the critical line; more precisely, $\kappa_0 > 0.0700162$.*

Remark 1.2. This proportion does not approach the strongest bounds available for $\zeta(s)$. The value of the present paper lies elsewhere: it gives a method for producing a positive proportion of zeros on the critical line for L -functions where the other methods may not be available, and it optimises the arithmetic on which that method rests.

This is proved in Section 11 as Theorem 11.5, by replacing Selberg's single mollifier $|M_U|^2$ by a positive-semidefinite sum of three squared mollifiers and optimising the resulting arithmetic quadratic form. The three-square detector is an explicit feasible point of that optimisation, built from the quarter-power logarithmic profile $v^{1/4}$; it is close to, but not, the optimum, and the true optimum lies only marginally higher. The stronger six-square profile of Appendix B already improves the constant only in its fifth decimal place, so we do not pursue the optimum here.

1.1 Selberg's method

Selberg's method rests on an elementary observation. On the critical line there is a real-valued function $Z(t)$, Hardy's function, whose real zeros are exactly the ordinates of the zeros of $\zeta(s)$ on the line. A sign change of $Z(t)$ between two ordinates forces a critical zero of odd order between them, so the number of sign changes on $[T, 2T]$ is a lower bound for the number of critical zeros there. Counted directly this bound is weak. Hardy and Hardy–Littlewood obtained in this way only $N_0(T) \gg T$ [9, 10], a density-zero set.

Selberg's device is to multiply $Z(t)$ by $|M(\frac{1}{2} + it)|^2$, where $M(s)$ is a Dirichlet polynomial called a mollifier. Since $|M(s)|^2 \geq 0$, the product keeps the sign changes of $Z(t)$ away from the zeros of $M(s)$, which are of even order and harmless, so it detects the same critical zeros. The gain is analytic, since the mollifier can be chosen to flatten the large fluctuations of $Z(t)$, sharpening the mean-value inequalities that turn average size into a count of sign changes. A second moment of the mollified detector, passed through Jensen's inequality, then yields a *positive* proportion of sign changes, so that $N_0(T) \gg T \log T$. How large a proportion is governed by the mean value of the mollified detector, hence by the coefficients of $M(s)$; choosing them well is the arithmetic problem at the heart of the method.

Zhuravlev made this quantitative for an arbitrary Dirichlet polynomial. We write U for the mollifier length, where Zhuravlev writes Z , so that $Z(t)$ is free for Hardy's function throughout. For real coefficients b_n , put

$$M_U(s) = \sum_{n \leq U} b_n n^{-s}, \quad Y_U(s) = \zeta(s) M_U(s) M_U(1-s).$$

Here $\aleph(s) = \chi(s)^{-1/2}$ is the unimodular factor defined in (9) below, where $\chi(s)$ is the factor in the functional equation $\zeta(s) = \chi(s) \zeta(1-s)$; it satisfies $|\aleph(\frac{1}{2} + it)| = 1$ and is what makes $\aleph(s) Y_U(s)$ real on the critical line. On the critical line, $\aleph(s) Y_U(s)$ is Hardy's function multiplied by $|M_U(s)|^2$, and so has the same sign changes as Hardy's function away from its zeros. Lavrik's approximate functional equation supplies an analytic half β_U with

$$\aleph(s) Y_U(s) = 2 \operatorname{Re} \beta_U(s) \quad (\operatorname{Re} s = \tfrac{1}{2}), \quad (1)$$

whose mean square is governed by

$$S_b(1, U) = \sum_{d \leq U^2} \varphi(d) \left(\sum_{\substack{\ell, m \leq U \\ d | \ell m}} \frac{b_\ell b_m}{\ell m} \right)^2. \quad (2)$$

If $U = T^\theta$, $0 < \theta < \frac{1}{4}$, and $S_b(1, U) \leq (\gamma + o(1))/\log U$, then under the normalisation $\aleph(s) Y_U(s) = c \operatorname{Re} \beta_U(s)$,

$$\kappa_0 \geq \frac{2\theta}{ec^2\gamma}. \quad (3)$$

Proposition 4.3 proves this, and Section 5 verifies its hypotheses uniformly in the coefficients used below.

Zhuravlev's coefficient choice gives $\gamma_{\text{Zh}} = 3\pi^2/16$. The source states (1) but its later logarithmic inequality is written as if the factor 2 were absent, and its polar term is printed with denominator $1-s$ in place of $s-1$. Correcting both points and carrying the missing term $-T \log 2$ through the optimisation gives the benchmark

$$\kappa_0 \geq \frac{1}{8e\gamma_{\text{Zh}}} = \frac{2}{3\pi^2 e} = 0.0248493202\dots, \quad (4)$$

rather than the printed proportion $2/21$.

1.2 Summary of the paper

We optimise the arithmetic form (2). Writing $\zeta(s)^{-1/2} = \sum_{n \geq 1} \lambda(n)n^{-s}$, the sharp coefficients $b_n = \lambda(n)$ give

$$S_\lambda(U) = \frac{\mathcal{C}_{\text{res}}}{\log U} + o\left(\frac{1}{\log U}\right), \quad \mathcal{C}_{\text{res}} = \frac{\Gamma(\frac{1}{4})^4}{2\pi^4} = \frac{2}{\Gamma(\frac{3}{4})^4}, \quad (5)$$

and hence

$$\kappa_0 \geq \frac{\Gamma(\frac{3}{4})^4}{16e} = 0.0518466520\dots \quad (6)$$

For the logarithmic power profiles $b_{n,a}(U) = \lambda(n)(\log(U/n)/\log U)^a$ with $a > 0$ (the name is ours; there seems to be no standard one), an explicit non-negative polyhedral constant $\mathcal{C}_a$ arises; at $a = \frac{1}{4}$, rigorous interval computation gives $0.6779054 < \mathcal{C}_{1/4} < 0.6786576$, and hence $\kappa_0 > 0.0677586$. Finally, allowing a positive-semidefinite sum of squared mollifiers turns the profile problem into a convex semidefinite programme, and the explicit three-square profile of Section 11 gives Theorem 1.1. Thus the sharp, quarter-power and three-square detectors improve the corrected Zhuravlev benchmark by factors of about 2.09, 2.73 and 2.82.

These factors are measured against Zhuravlev's benchmark, which rests on Wirsing's mean-value theorem; his coefficients, evaluated exactly rather than bounded, already give $0.8052\dots$ (Section 10.1). Most of the improvement therefore lies in evaluating the diagonal rather than in the choice of coefficients, and what this paper adds to the arithmetic is the profile: the multiplicative structure of λ is already optimal, by the local inequality at the end of Section 11.

The proof separates into analytic and arithmetic parts. The analytic reconstruction uses the sign-preserving Selberg detector, Lavrik's approximate functional equation, a rectangle inequality, and a coefficient-uniform rational-frequency off-diagonal estimate. The arithmetic analysis begins with the convolution identity $\lambda * \lambda = \mu$, removes the same-side poles from a four-variable Euler product, and evaluates the resulting bipartite quarter-order residue exactly; the positive-semidefinite optimisation is then a finite convex programme certified in Appendix B.

Section 2 gives the historical comparison. Sections 3–5 reconstruct and correct Zhuravlev's analytic argument. Sections 6–9 evaluate the sharp diagonal, Section 10 treats the power profiles, Section 11 carries out the positive-semidefinite optimisation, with the resulting critical-line proportions at its end. Appendix A supplies the analytic large-disk bounds and Appendix B the finite-dimensional certificates.

2 Historical context: Hardy, Selberg, Levinson, Atkinson and pair correlation

Before giving the full details of the argument, we explore the various methods that have been used to produce values of κ_0 , and where the present method sits among them.

2.1 Hardy's function and sign changes

Write the functional equation of $\zeta(s)$ as

$$\zeta(s) = \chi(s)\zeta(1-s), \quad \chi(s) = \pi^{s-1/2} \frac{\Gamma((1-s)/2)}{\Gamma(s/2)},$$

and choose a continuous square root on the critical line. Hardy's function

$$Z(t) = \chi(\tfrac{1}{2} + it)^{-1/2} \zeta(\tfrac{1}{2} + it)$$

is real for real t , and its real zeros are precisely the ordinates of zeros of $\zeta(s)$ on $\text{Re } s = \frac{1}{2}$. Every sign change therefore detects a zero of odd multiplicity. Hardy's theorem and the Hardy–Littlewood lower bound are based on this principle, supplemented by mean-value estimates

for $Z(t)$ [9, 10, 19]. It is worth recalling how the principle is applied, since the same shape of argument recurs throughout. Hardy’s original argument compares two averages of $Z(t)$ over $[T, 2T]$. The mean $\int_T^{2T} Z(t) dt$ is small, because $Z(t)$ oscillates, while the mean of $|Z(t)|$, bounded below through $\int_T^{2T} Z(t)^2 dt \asymp T \log T$ and Cauchy–Schwarz, is not. If $Z(t)$ had constant sign on a long subinterval the two averages would agree there in size, so $Z(t)$ must change sign, and a positive number of times. Hardy–Littlewood turned this into the quantitative count $N_0(T) \gg T$ by running the comparison on each of $\asymp T$ short intervals. The whole difficulty is that the second moment, which supplies the lower bound, is dominated by the rare large values of $Z(t)$ rather than by its typical size, so the comparison is lossy; Selberg’s mollifier is precisely a device for flattening those large values before the comparison is made. A sign-change count is at most $N_0(T)$, but it has the additional feature that it counts only zeros of odd multiplicity.

2.2 Selberg’s sign-preserving mollifier

Selberg multiplied the real detector by a non-negative mollifier. If $M(s)$ is a Dirichlet polynomial, then on the critical line

$$\chi(s)^{-1/2} \zeta(s) M(s) M(1-s) = Z(t) |M(\tfrac{1}{2} + it)|^2.$$

Zeros of $M(s)$ occur with even order in this real function and do not create false sign changes. A mollifier may therefore reduce the mean size of the detector without changing its sign. Selberg used the coefficients of $\zeta(s)^{-1/2}$, with a linear logarithmic weight, and combined the mollified mean square with a Littlewood-type rectangle argument to prove a positive proportion of critical-line zeros [18, 19].

The first explicit numerical evaluation of Selberg’s implicit constant is usually credited to S.-H. Min. Titchmarsh records that the constant was calculated in Min’s 1947 Oxford D.Phil. dissertation [19, 16, §10.9, pp. 268–269]. Secondary sources report $1/14,074,731$ for the dissertation and $1/60,000$ for Min’s later paper in the *Journal of Peking University* [21]. Since neither original has been inspected by the author, we record these two numerical attributions as secondary-source reports.

The quantitative problem splits into three parts: a sign-change inequality, a mean-value theorem for an analytic half of the mollified functional equation, and an arithmetic optimisation of the mollifier coefficients. Zhuravlev’s paper is a particularly explicit implementation of this scheme. It allows general initial coefficients, reduces the mean square to the form $S_b(1, U)$, and then chooses one multiplicative sequence to obtain a numerical constant.

2.3 Levinson’s argument-principle method

Levinson introduced a different zero detector. One forms a differential combination such as

$$G(s) = \zeta(s) + \frac{\zeta'(s)}{\log T},$$

or, more generally, $V(s) = Q(-\log(T)^{-1} d/ds) \zeta(s)$, and applies the argument principle on a line $\sigma_0 = \frac{1}{2} - R/\log T$. A mollifier approximating $1/\zeta(s)$ is then used to estimate a twisted second moment of $V(s)$. This counts zeros through complex-analytic variation on a displaced line, rather than through sign changes of a real function. In more detail, the point of the differential combination is that $V(s)$ inherits the zeros of $\zeta(s)$ that lie on the critical line but has its own zeros pushed off it in a controlled way. Counting the zeros of $V(s)$ in the rectangle $\sigma_0 < \Re s < \frac{1}{2}$, $T < \Im s < 2T$ by the argument principle, and converting that count by Littlewood’s lemma, replaces it by $\frac{1}{2\pi} \int_T^{2T} \log |V(\sigma_0 + it)| dt$ up to a term the functional equation supplies exactly. Every zero of $\zeta(s)$ off the critical line contributes to the rectangle count, so an upper bound for the integral is a lower bound for the proportion on the line. The integral is estimated by

Jensen's inequality from the mollified second moment $\int_T^{2T} |V(\sigma_0 + it)\psi(\sigma_0 + it)|^2 dt$, where $\psi(s)$ is a Dirichlet polynomial approximating $\zeta(s)^{-1}$; the mollifier is what keeps that mean square bounded, and its length is what limits the resulting proportion.

Levinson proved that more than one third of the zeros lie on the critical line [15]. Conrey obtained more than two fifths [5], and later multi-piece and higher-order mollifiers pushed the proportion beyond 41% [4]. The strongest bound from the mollifier methods is the theorem of Pratt, Robles, Zaharescu and Zeindler that more than five twelfths of the zeros lie on the line [17].

2.4 Atkinson's two-point detector

Atkinson's 1948 argument is a third method [2]. It uses Hardy's real function at two nearby points. The displacement is chosen through the Hardy phase, and the inequality

$$Z(t)Z(t') < 0$$

forces a zero between t and t' . The measure of the negative set is controlled by a shifted mixed mean of $Z(t)Z(t')$ and by Ingham's fourth moment [2, 11]. Baluyot later combined this detector with a mollifier and a twisted fourth-moment formula to give another proof of a positive proportion [3]. His Theorem 1.1 is stated qualitatively, but the closing section of that paper optimises the argument explicitly and gives $\kappa_0 > 0.0001049$, rising to $\kappa_0 > 0.0086729$ if his fourth-moment estimates are granted for mollifiers of any length.

Selberg and Atkinson are both sign-change methods, but their statistics are different: Selberg uses a one-point mollified second moment, whereas Atkinson uses a two-point correlation and a fourth-moment majorant. Levinson's method is different from both, since it applies the argument principle to a mollified differential combination off the critical line.

2.5 The pair-correlation method

The three methods above are local. Each forms a mollified detector, a Dirichlet polynomial times $\zeta(s)$ or a differential combination of $\zeta(s)$, and reads critical zeros off a first, second or fourth moment of that detector. A fourth route is global, and works through the pair correlation of the zeros. Montgomery's 1973 study of the pair correlation, via Weil's explicit formula, showed under the Riemann hypothesis that at least two thirds of the zeros are simple. The explicit formula supplies a Hermitian form

$$W(f, g) = \sum_{\rho} m_{\rho} \hat{f}(\gamma_{\rho}) \hat{g}(\gamma_{\rho})$$

over the zeros, whose positivity on every test function is equivalent to the Riemann hypothesis. Montgomery's evaluation of the associated prime-side second moment is a mean value of a Dirichlet polynomial of length T and is unconditional; the hypothesis entered only to read the zero side termwise as a positive sum over real ordinates.

That last step was made unconditional in 2026, raising the record for the proportion of zeros on the critical line to two thirds [1]. The idea is to replace the single Hermitian form W by a finite matrix. One takes a family of about $N(T, 2T)$ modulated copies of one fixed window, equispaced through $[T, 2T]$, and assembles the values of the explicit formula on them into a finite real symmetric matrix. On the zero side the matrix is a sum of two pieces: each zero actually on the critical line contributes a positive rank-one piece, while each pair of zeros off the line contributes a piece with one positive and one negative direction. Hence a matrix with few positive directions can accommodate only few on-line zeros, and the classical inertia and rank-trace inequalities of linear algebra make this quantitative. On the prime side the same matrix has a size that Montgomery's mean value pins down unconditionally. Comparing the two sides bounds the number of on-line zeros from below, and the strength of the bound depends only

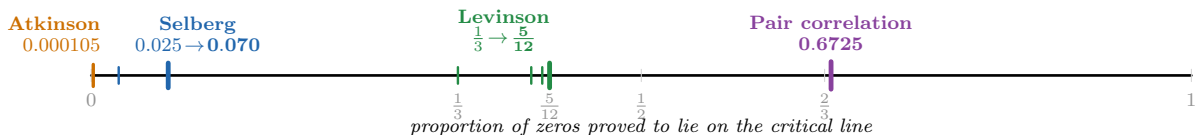


Figure 1: Lower bounds on the proportion of nontrivial zeros proved to lie on the critical line, shown on a single axis for comparison. Each method appears in its own colour, with its current record in bold and earlier records as thin ticks. Atkinson’s constant is explicit but very small, so its tick sits at the origin at this scale; Baluyot’s optimisation of the constant in Atkinson’s method gives 0.000105, rising to 0.0087 if his fourth-moment input is granted for mollifiers of any length.

on the choice of window: the indicator window gives two thirds, and the Montgomery–Taylor window gives 0.6725. What matters for us is that the positivity is used globally, over the whole family of zeros at once, rather than locally through a detector.

Shortly after, Lamzouri [13] obtained the same bound by a shorter argument. Replacing the finite-dimensional matrix framework above by a single inequality in the Hilbert space of the explicit formula, he applies Montgomery’s pair-correlation theorem directly.

3 Zhuravlev’s argument and the corrections required

Unlike Levinson and Atkinson, Zhuravlev did not introduce a new detector. His contribution was to carry Selberg’s scheme through to an explicit constant for a general Dirichlet polynomial, by the analytic-half and mean-value steps outlined in Section 1.1. This section and the two that follow reconstruct that implementation, identify the normalisation discrepancy behind the corrected benchmark, and prove the coefficient-uniform estimates needed for the corrected theorem. We follow Zhuravlev’s four-stage architecture, but distinguish quoted inputs from corrected or independently reproved statements.

3.1 Structure of the argument

For a Dirichlet polynomial $M_U(s)$, form $Y_U(s) = \zeta(s)M_U(s)M_U(1-s)$. On the critical line its sign is that of Hardy’s function. Lavrik’s approximate functional equation splits $Y_U(s)$ into two conjugate halves and defines an analytic half $\Lambda_U(s)$, a Dirichlet series in the generalised frequencies of $Y_U(s)$, of which $Y_U(s)$ is twice the real part after multiplication by the unimodular factor $\aleph(s)$; we write $\beta_U(s) = \aleph(s)\Lambda_U(s)$ for that rotated half, so that $\aleph(s)Y_U(s) = 2 \operatorname{Re} \beta_U(s)$ on the critical line. A rectangle form of Littlewood’s lemma, followed by Jensen’s inequality, bounds sign changes in terms of the second moment of $\beta_U(s)$ on $\operatorname{Re} s = \frac{1}{2} - x/2$, where $x = 1 - 2\delta$ measures how far that line has been displaced to the left of the critical line. The diagonal part of this moment is then expressed through $S_b(1, U)$. The first three stages are analytic; the last is the arithmetic optimisation addressed in the second half of this paper.

3.2 Zhuravlev’s coefficient choice

Zhuravlev introduces

$$\beta_d = \frac{f(d)}{N}, \quad N = \sum_{d \leq U} \mu(d)f(d),$$

where f is squarefree-supported and multiplicative. For primes larger than 10, $f(p) = -1/(2(p-1))$, while

$$f(2) = -\frac{1}{4}, \quad f(3) = -\frac{17}{100}, \quad f(5) = -\frac{1}{10}, \quad f(7) = -\frac{7}{100}.$$

The coefficients b_ℓ are recovered from

$$\frac{b_\ell}{\ell} = \sum_{d \leq U} \mu\left(\frac{d}{\ell}\right) \beta_d.$$

We shall need the following elementary bounds in the analytic argument. Because f is squarefree-supported, is negative at every prime, and is multiplicative,

$$\mu(d)f(d) = |f(d)|, \quad N = \sum_{d \leq U} |f(d)|, \quad 1 \leq N \leq U.$$

If ℓ is not squarefree, then $b_\ell = 0$. If ℓ is squarefree, write $d = \ell q$ in the preceding inversion formula. Then

$$\frac{b_\ell}{\ell} = \frac{f(\ell)}{N} \sum_{\substack{q \leq U/\ell \\ (q, \ell)=1}} |f(q)|, \quad |b_\ell| \leq \ell |f(\ell)| = \prod_{p|\ell} p |f(p)| \leq 1.$$

Here $p|f(p)| \leq 0.51$ for $p = 2, 3, 5, 7$, while $p|f(p)| = p/(2(p-1)) \leq 11/20$ for $p \geq 11$. In particular, if $U > 10$ is prime, only $d = U$ contributes to the terminal coefficient and

$$\frac{b_U}{U} = \frac{f(U)}{N} = -\frac{1}{2(U-1)N}, \quad |b_U| \geq \frac{1}{2U}, \quad \log^+ \frac{1}{|b_U|} \ll \log U. \quad (7)$$

A Möbius decomposition gives

$$S_b(1, U) \leq N^{-4} \sum_{d \leq U^2} \varphi(d) F(d)^2,$$

where $\varphi(d)F(d)^2$ is multiplicative and, with $u_p = |f(p)|$,

$$\begin{aligned} \varphi(p)F(p)^2 &= (p-1)(2u_p + 3u_p^2)^2, \\ \varphi(p^2)F(p^2)^2 &= p(p-1)u_p^4, \end{aligned}$$

with no higher prime-power terms.

Proposition 3.1. *The preceding coefficients satisfy*

$$S_b(1, U) \leq \left(\frac{3\pi^2}{16} + o(1) \right) \frac{1}{\log U}. \quad (8)$$

Proof. Applying Wirsing's mean-value theorem [20] with exponents $1/2$ and 1 , respectively, gives

$$S_b(1, U) \leq \frac{\pi^2}{8 \log U} \left\{ \prod_p R_p(u_p) + o(1) \right\},$$

where

$$R_p(u) = \frac{1 + (p-1)(2u + 3u^2)^2 + p(p-1)u^4}{(1+u)^4(1-p^{-1})}.$$

For $p > 10$, direct simplification at $u = 1/(2(p-1))$ gives

$$R_p(u) \leq 1 + \frac{17}{16(p-1)^2}.$$

A finite rational calculation gives

$$\prod_{p \in \{2, 3, 5, 7\}} R_p(u_p) \prod_{11 \leq p \leq 31} \left(1 + \frac{17}{16(p-1)^2} \right) < 1.446.$$

For the remaining primes,

$$\log \prod_{p>31} \left(1 + \frac{17}{16(p-1)^2}\right) \leq \frac{17}{16} \sum_{n \geq 32} \frac{1}{(n-1)^2} < \frac{17}{480},$$

so the tail is less than $e^{17/480} < 1.037$. Hence $\prod_p R_p(u_p) < 3/2$, which proves (8). $\square$

The shape of these coefficients is not arbitrary, and neither is the loss they appear to incur. Section 10.1 shows that $pf(p) \rightarrow -\frac{1}{2}$ is forced by the power of $\log U$; that the restriction to squarefree support costs under 5%; that Zhuravlev's normalisation N silently supplies a logarithmic profile which is worth a factor 0.81; and that his coefficients, evaluated by the machinery of Sections 6–10 instead of by Wirsing's inequality, already give $\kappa_0 \geq 0.057$. Most of the distance between (8) and the results below is therefore in the mean-value estimate rather than in the coefficients, which are close to the best available in their class.

3.3 Corrections to the printed argument

There are three issues relevant to the proof.

- (i) The source states $\aleph(s)Y_U(s) = 2 \operatorname{Re} \beta_U(s)$, but its later logarithmic inequality is written as if the factor 2 were absent. The missing term is $-T \log 2$. At fixed arithmetic input this divides the exact unrounded output $8/(3\pi^2 e)$ by four and gives $2/(3\pi^2 e)$. Zhuravlev subsequently rounded the uncorrected output down to $2/21$; consequently the corrected exact value is not literally $(2/21)/4$.
- (ii) The secondary term in the printed truncated-diagonal formula is not the coefficient-uniform identity for arbitrary b_n . Direct expansion gives the non-negative quadratic form $Q_b(2\delta, U)$ in Lemma 5.5. Its coefficient is $\zeta(2\delta) < 0$, which is exactly what the upper bound requires.
- (iii) Several exponents and an integration endpoint in the printed off-diagonal formula are indistinct. We therefore do not reconstruct a theorem by conjecturally correcting that display. Proposition 5.8 uses instead an independent spacing argument for the rational frequencies.

The original copy also resolves a number of local typographical readings, including the denominator $\Gamma(s/2)(s-1)$ in Lavrik's polar term and the prime $p = U$ in the horizontal Jensen argument. These local corrections do not alter the leading constant.

There is one further editorial correction to the electronic source material that we have of the original paper. In the facsimile on original p. 44, and in Lavrik's defining integral, the kernel is X^{-z} . An earlier electronic transcription and translation omitted the minus sign and displayed X^z . We use the facsimile reading X^{-z} below. This is a transcription error, not an error in Zhuravlev's printed paper.

3.4 Original and modern ingredients

The zero detector, Littlewood's rectangle argument, Lavrik's approximate functional equation, Wirsing's theorem, and direct mean-value expansions were all available in Zhuravlev's period. Our reconstruction changes neither the detector nor its conceptual basis. The modern elements are the normalisation-invariant formulation (3), the coefficient-uniform rational-frequency proof of the off-diagonal estimate (Section 5), the four-variable Euler-product and polyhedral analysis (Sections 7–8), the uniform complex-power Selberg–Delange transfer (Section 9), and the rigorous interval certification of the weighted constant (Section 10 and Appendix B). These tools make the method easier to verify and expose the arithmetic optimisation problem more clearly, but they should not be confused with a new zero-detection principle.

4 Zhuravlev's sign-change reduction

We now give the part of Zhuravlev's argument that converts a mollified mean-square estimate into critical-line zeros. This section is independent of our later choice of coefficients.

Let

$$M_U(s) = \sum_{n \leq U} b_n n^{-s}, \quad b_n \in \mathbb{R}, \quad b_1 = 1, \quad |b_n| \leq 1.$$

Also put

$$Y_U(s) = \zeta(s) M_U(s) M_U(1-s).$$

Since the two mollifier factors are interchanged by $s \mapsto 1-s$, Y_U has the same functional equation as ζ :

$$Y_U(s) = \chi(s) Y_U(1-s).$$

Set

$$\aleph(s) = \chi(s)^{-1/2} = \left\{ \pi^{1/2-s} \frac{\Gamma(s/2)}{\Gamma((1-s)/2)} \right\}^{1/2}, \quad (9)$$

with the branch chosen continuously in the region under consideration. On $\operatorname{Re} s = \frac{1}{2}$ one has $|\aleph(s)| = 1$ and

$$\aleph(s) Y_U(s) = \aleph(s) \zeta(s) |M_U(s)|^2 \in \mathbb{R}.$$

Consequently its sign changes occur only at zeros of ζ of odd multiplicity; zeros of M_U enter through $|M_U|^2$ and have even order.

Expand Y_U as a generalised Dirichlet series

$$Y_U(s) = \sum_{\xi > 0} a_\xi \xi^{-s},$$

where the frequencies are positive rationals. Lavrik's symmetric approximate functional equation, in the form used by Zhuravlev, constructs an analytic function $\Lambda_U(s)$ representing one half of this functional equation. We write

$$\beta_U(s) = \aleph(s) \Lambda_U(s).$$

The two properties needed below are

$$\aleph(s) Y_U(s) = 2 \operatorname{Re} \beta_U(s) \quad (\operatorname{Re} s = \tfrac{1}{2}) \quad (10)$$

and, for $s = \sigma + it$ with $t \asymp T$,

$$\Lambda_U(s) = \sum_{\xi < \sqrt{t/(2\pi)}} a_\xi \xi^{-s} + O\left(t^{-\sigma/2} U \log^3 T\right). \quad (11)$$

Section 5 states and proves the uniform form of (11), together with the horizontal argument bounds and the off-diagonal estimate. Here we show how these inputs are used.

Let $A(T, 2T)$ denote the number of distinct ordinates in $[T, 2T]$ at which $\zeta(\frac{1}{2} + it)$ has a zero of odd multiplicity. Then $A(T, 2T) \leq N_0(2T) - N_0(T)$.

Lemma 4.1. *Let $0 < \delta < \frac{1}{2}$, put $x = 1 - 2\delta$, and suppose that*

$$\aleph(s) Y_U(s) = c \operatorname{Re} \beta_U(s) \quad (\operatorname{Re} s = \tfrac{1}{2}), \quad c > 0.$$

Assume that the net horizontal argument increments of β_U in the rectangle $\delta \leq \operatorname{Re} s \leq \frac{1}{2}$, $T \leq \operatorname{Im} s \leq 2T$, and the corresponding auxiliary boundary increments, contribute $o(T)$. Then

$$\frac{\pi x}{2} A(T, 2T) \geq - \int_T^{2T} \log |\beta_U(\delta + it)| dt - T \log c - o(T). \quad (12)$$

Proof. For a function h analytic in a rectangle $\sigma_1 \leq \operatorname{Re} s \leq \sigma_2$, $T \leq \operatorname{Im} s \leq 2T$, Littlewood's rectangle argument gives

$$\begin{aligned} & \int_T^{2T} \log |h(\sigma_1 + it)| dt + \int_{\sigma_1}^{\sigma_2} \arg h(\sigma + 2iT) d\sigma \\ & \geq \int_T^{2T} \log |h(\sigma_2 + it)| dt + \int_{\sigma_1}^{\sigma_2} \arg h(\sigma + iT) d\sigma, \end{aligned} \quad (13)$$

with the usual limiting convention if a zero lies on the boundary.

On the critical line, the hypothesis of the lemma gives

$$|\beta_U(\tfrac{1}{2} + it)| \geq |\operatorname{Re} \beta_U(\tfrac{1}{2} + it)| = \frac{1}{c} |\aleph(\tfrac{1}{2} + it) Y_U(\tfrac{1}{2} + it)|.$$

Since $|\aleph| = 1$ there,

$$\int_T^{2T} \log |\beta_U(\tfrac{1}{2} + it)| dt \geq \int_T^{2T} \log |\zeta(\tfrac{1}{2} + it)| dt + 2 \int_T^{2T} \log |M_U(\tfrac{1}{2} + it)| dt - T \log c. \quad (14)$$

Apply (13) separately to ζ and M_U in the rectangle $\frac{1}{2} \leq \operatorname{Re} s \leq 3$. On the right edge, $|\zeta(3 + it) - 1|$ and $|M_U(3 + it) - 1|$ are both at most $\zeta(3) - 1 < 1$. Their principal logarithms therefore have absolutely convergent Dirichlet expansions with zero constant term, so their vertical integrals are $O(\log T)$ and $O(1)$. The corresponding horizontal argument increments are $O(\log T)$ for ζ and $O(U)$ for M_U : the real and imaginary parts of $M_U(\sigma + it)$ are exponential polynomials in σ with at most U terms. It follows from (14) that

$$\int_T^{2T} \log |\beta_U(\tfrac{1}{2} + it)| dt \geq -T \log c - o(T). \quad (15)$$

It remains to control the horizontal boundary terms when (13) is applied to $h = \beta_U$ between δ and $\frac{1}{2}$. Along the critical edge, the real part of β_U is the real sign-detecting function $\aleph(s) Y_U(s)$. Divide $[T, 2T]$ at the sign changes of this real part and choose the argument continuously from the lower endpoint, with the usual infinitesimal detours around zeros. On each intervening interval the curve $\beta_U(\tfrac{1}{2} + it)$ remains in one closed half-plane, so its argument can change by at most π . Consequently, for every $u \in [T, 2T]$,

$$|\arg \beta_U(\tfrac{1}{2} + iu) - \arg \beta_U(\tfrac{1}{2} + iT)| \leq \pi \{A(T, u) + 1\}.$$

Zeros of $M_U(s)$ do not create additional sign changes, because they occur in $\aleph(s) Y_U(s) = Z(t) |M_U(s)|^2$ with even order. Transporting this common branch from the critical edge to the two horizontal edges changes it by only the assumed $o(T)$ boundary increments. Hence the difference of the two horizontal argument integrals in (13) is at most $\pi(\frac{1}{2} - \delta)A(T, 2T) + o(T)$. Using this and (15) gives

$$\pi \left(\frac{1}{2} - \delta \right) A(T, 2T) \geq - \int_T^{2T} \log |\beta_U(\delta + it)| dt - T \log c - o(T),$$

which is (12) because $\frac{1}{2} - \delta = x/2$. □

Zhuravlev's arithmetic form is

$$S_b(1, U) = \sum_{d \leq U^2} \varphi(d) \left(\sum_{\substack{\ell, m \leq U \\ d | \ell m}} \frac{b_\ell b_m}{\ell m} \right)^2. \quad (16)$$

The following elementary lemma records the passage from dyadic intervals to the global lower proportion.

Lemma 4.2. *Suppose that, for some $q \geq 0$,*

$$N_0(2T) - N_0(T) \geq q\{N(2T) - N(T)\} + o(T \log T)$$

as $T \rightarrow \infty$. Then $\kappa_0 \geq q$.

Proof. Set $D(X) = N_0(X) - qN(X)$. Applying the hypothesis with $T = X/2$ gives

$$D(X) \geq D(X/2) - \varepsilon(X)X \log X, \quad \varepsilon(X) \rightarrow 0.$$

Iteration down to a fixed height yields

$$D(X) \geq - \sum_{j \geq 0} \varepsilon(X/2^j) \frac{X}{2^j} \log X + O(1).$$

For any fixed J , the first J terms are $o(X \log X)$; the remaining geometric tail is $O(2^{-J} X \log X)$. Letting first $X \rightarrow \infty$ and then $J \rightarrow \infty$ gives $D(X) = o_-(X \log X)$. Since $N(X) \sim X \log X / (2\pi)$, the result follows. $\square$

The next proposition isolates the optimisation step at the heart of the method.

Proposition 4.3. *Let $U \asymp T^\theta$, $\log U = \theta \log T + O(1)$, $0 < \theta < 1/4$, and $x = 1 - 2\delta \asymp 1/\log T$. Suppose*

$$\Re(s)Y_U(s) = c \Re \beta_U(s) \quad (\Re s = \tfrac{1}{2}), \quad c > 0,$$

the horizontal errors in Lemma 4.1 are $o(T)$, and, uniformly in this moving range,

$$\frac{1}{T} \int_T^{2T} |\beta_U(\delta + it)|^2 dt \leq \frac{\gamma}{x \log U} (1 + o(1)). \quad (17)$$

Then

$$A(T, 2T) \geq \frac{T \log U}{\pi e c^2 \gamma} (1 + o(1)), \quad \kappa_0 \geq \frac{2\theta}{e c^2 \gamma}.$$

Proof. By concavity of the logarithm and Cauchy's inequality,

$$\int_T^{2T} \log |\beta_U(\delta + it)| dt \leq \frac{T}{2} \log \left(\frac{1}{T} \int_T^{2T} |\beta_U(\delta + it)|^2 dt \right).$$

The critical-line identity contributes $-T \log c$ to the rectangle lemma, so using (17) gives

$$A(T, 2T) \geq \frac{T}{\pi x} \left\{ \log \frac{x \log U}{\gamma} - 2 \log c \right\} + o(T/x).$$

Put $u = x \log U / \gamma$. The function $\log(u/c^2)/u$ is maximised at $u = e c^2$. Taking

$$x = \frac{e c^2 \gamma}{\log U}$$

therefore yields

$$A(T, 2T) \geq \frac{T \log U}{\pi e c^2 \gamma} (1 + o(1)).$$

Finally,

$$N(2T) - N(T) \sim \frac{T}{2\pi} \log T, \quad \log U = \theta \log T,$$

so the same lower proportion holds on every sufficiently large dyadic interval. Lemma 4.2, with $q = 2\theta/(e c^2 \gamma)$, now gives

$$\liminf_{T \rightarrow \infty} \frac{N_0(T)}{N(T)} \geq \frac{2\theta}{e c^2 \gamma}.$$

$\square$

The original Russian source has $c = 2$. Its equation (5) omits the resulting term $-T \log 2$ and consequently optimises as though $c = 1$. The remainder of the paper computes improved arithmetic constants γ and verifies the analytic hypotheses independently of this correction.

5 Coefficient-uniform analytic estimates

This section verifies the analytic input required by Proposition 4.3. We take Lavrik's general theorem on analytic halves of Dirichlet functional equations as a quoted foundational input, and derive from it the coefficient-uniform approximate functional equation and the horizontal Jensen bound used by Zhuravlev. For the mean square we give a new proof of the off-diagonal estimate, based only on the spacing of the rational frequencies. This avoids any dependence on the typographically ambiguous exponents in Zhuravlev's printed equation (8), whose exact reading is not needed for the reconstructed proof.

Let

$$a_\xi = \sum_{\substack{n \geq 1, \ell, m \leq U \\ n\ell/m = \xi}} \frac{b_\ell b_m}{m}, \quad Y_U(s) = \sum_{\xi > 0} a_\xi \xi^{-s} \quad (\operatorname{Re} s > 1), \quad (18)$$

We now make the analytic half, and its dependence on the smoothing parameter, explicit. On a dyadic block $T \leq \operatorname{Im} s \leq 2T$, fix

$$\rho_T^+ = \exp\left\{i\left(\frac{\pi}{2} - \frac{1}{T}\right)\right\}, \quad \rho_T^- = \overline{\rho_T^+}.$$

These parameters remain fixed while s moves in the rectangle; this point is needed for analyticity. For $|\arg X^2| < \pi/2$, put

$$J(s, X) = \frac{1}{2\pi i} \int_{\operatorname{Re} z = \Delta} \Gamma\left(\frac{s+z}{2}\right) \frac{X^{-z}}{z} dz, \quad \Delta > \max(0, \operatorname{Re} s), \quad (19)$$

$$F(s, X) = \frac{J(s, X)}{\Gamma(s/2)} = \frac{\Gamma(s/2, X^2)}{\Gamma(s/2)}. \quad (20)$$

The last equality follows first for $\operatorname{Re} s > 0$ by differentiating in X , and elsewhere by analytic continuation. Notice the minus sign in X^{-z} ; it is visible in both Lavrik's formula and Zhuravlev's facsimile. Define the paired analytic halves

$$\Lambda_U^\pm(s) = \sum_{\xi > 0} a_\xi \xi^{-s} F\left(s, (\pi \rho_T^\pm)^{1/2} \xi\right) + \frac{\pi^{1/2} (\pi \rho_T^\pm)^{(s-1)/2} M_U(0) M_U(1)}{\Gamma(s/2)(s-1)}. \quad (21)$$

We write $\Lambda_U = \Lambda_U^+$ and $\beta_U(s) = \aleph(s) \Lambda_U^+(s)$ on the positive block. Since the coefficients are real,

$$\Lambda_U^-(\sigma - iu) = \overline{\Lambda_U^+(\sigma + iu)}.$$

Lavrik's fundamental identity [14, Lemma 2.1, English p. 133], specialised to the functional equation of Y_U , gives

$$\aleph(s) Y_U(s) = 2 \operatorname{Re}\{\aleph(s) \Lambda_U^+(s)\} \quad (\operatorname{Re} s = \tfrac{1}{2}, T \leq \operatorname{Im} s \leq 2T).$$

Lavrik's Theorem 1 [14, English pp. 130–131] is applied below with its required ordinate-dependent phase. Lemma 5.2 compares that pointwise half with the fixed analytic half above. Appendix A derives the larger-disk bounds required for Jensen directly from (20); no fixed-strip theorem is used outside its stated range.

Put

$$R_b(U) = \left(\sum_{\ell, m \leq U} \frac{|b_\ell b_m|}{m} \right)^2. \quad (22)$$

If $|b_n| \leq 1$, then

$$R_b(U) \ll U^2 \log^2(2U). \quad (23)$$

Lemma 5.1. *Uniformly for $A \geq 2$ and $h > 0$,*

$$\sum_{n \geq 1} \frac{\exp\{-(hn)^2/A^2\}}{1 + |A - \pi(hn)^2/A|} \ll (1 + h^{-1}) \log(2A).$$

Proof. Put $y_0 = A/\sqrt{\pi}$. On $hn \leq y_0/2$ and $hn \geq 2y_0$ the denominator is $\gg A + (hn)^2/A$, and comparison with an integral gives $O(1 + h^{-1})$. In the remaining range,

$$1 + |A - \pi(hn)^2/A| \asymp 1 + |hn - y_0|.$$

Divide this range into the sets $j \leq |hn - y_0| < j + 1$. Each contains $O(1 + h^{-1})$ integers n , and summing $(1 + j)^{-1}$ for $j \ll A$ gives the stated logarithm. $\square$

Lemma 5.2. *Let $s = \sigma + it$, where $T \leq t \leq 2T$ and $0 \leq \sigma \leq 1/2$, and put*

$$\rho(t) = \exp\left\{i\left(\frac{\pi}{2} - \frac{1}{t}\right)\right\}.$$

Let $\tilde{\Lambda}_{U,t}(s)$ be the expression in (21) with ρ_T^+ replaced by $\rho(t)$. If $b_1 = 1$ and $|b_n| \leq 1$ for $n \leq U$, then

$$\Lambda_U(s) - \tilde{\Lambda}_{U,t}(s) \ll t^{-\sigma/2} U \log(2U), \quad (24)$$

uniformly in the displayed range.

Proof. Write $a = s/2$, $r = \pi\xi^2$, $\phi_T = \pi/2 - 1/T$, and $\phi_t = \pi/2 - 1/t$. The two incomplete-gamma endpoints $re^{i\phi_T}$ and $re^{i\phi_t}$ are joined by the circular arc $\mathcal{A}_r$. Hence

$$F(s, (\pi\rho_T^+)^{1/2}\xi) - F(s, (\pi\rho(t))^{1/2}\xi) = -\frac{1}{\Gamma(a)} \int_{\mathcal{A}_r} e^{-u} u^{a-1} du.$$

On this arc,

$$|\phi_t - \phi_T| \leq T^{-1}, \quad \cos(\arg u) \gg t^{-1}, \quad \exp\left\{t\left(\frac{\pi}{4} - \frac{\arg u}{2}\right)\right\} \ll 1.$$

Uniform Stirling in $0 \leq \sigma \leq 1/2$ therefore gives

$$\left|F(s, (\pi\rho_T^+)^{1/2}\xi) - F(s, (\pi\rho(t))^{1/2}\xi)\right| \ll t^{-1/2-\sigma/2} \xi^\sigma e^{-c\xi^2/t}. \quad (25)$$

Indeed, the arc length is $r|\phi_t - \phi_T|$, while $|u^{a-1}| = r^{\sigma/2-1} e^{-t \arg u/2}$ and

$$|\Gamma(a)|^{-1} \ll t^{1/2-\sigma/2} e^{\pi t/4}.$$

Expanding a_ξ by (18) and using $\sum_{n \geq 1} e^{-An^2} \ll 1 + A^{-1/2}$, we obtain

$$\begin{aligned} \sum_{\xi > 0} |a_\xi| e^{-c\xi^2/t} &\leq \sum_{\ell, m \leq U} \frac{1}{m} \sum_{n \geq 1} e^{-c(n\ell/m)^2/t} \\ &\ll \sum_{\ell, m \leq U} \left(\frac{1}{m} + \frac{\sqrt{t}}{\ell}\right) \ll U\sqrt{t} \log(2U). \end{aligned}$$

Thus (25), after multiplication by $\xi^{-\sigma}$, contributes $O(t^{-\sigma/2} U \log(2U))$.

It remains to compare the polar terms. On the interval between ϕ_T and ϕ_t , differentiation of $e^{i\phi(s-1)/2}$, followed by the same Stirling estimate and $|M_U(0)M_U(1)| \ll U \log(2U)$, gives

$$\frac{\sqrt{\pi} |M_U(0)M_U(1)|}{|\Gamma(s/2)(s-1)|} \left|e^{i\phi_T(s-1)/2} - e^{i\phi_t(s-1)/2}\right| \ll t^{-1/2-\sigma/2} U \log(2U).$$

This is smaller than the asserted bound and proves the lemma. $\square$

Lemma 5.3. *Let $b_n \in \mathbb{R}$, $b_1 = 1$, and $|b_n| \leq 1$ for $n \leq U$. Let $T \leq t \leq 2T$, $0 \leq \sigma \leq 1/2$, and $U < T^{1/4}$. With*

$$\tau(t) = \sqrt{t/(2\pi)}, \quad \Lambda_U^0(s; t) = \sum_{\xi < \tau(t)} a_\xi \xi^{-s},$$

Lavrik's approximate-functional-equation function satisfies

$$\Lambda_U(s) = \Lambda_U^0(s; t) + O\left(t^{-\sigma/2} U \log^3(2T)\right), \quad (26)$$

uniformly in the displayed range. Consequently, if $\beta_U(s) = \aleph(s) \Lambda_U(s)$ and $\delta = \operatorname{Re} s$, then

$$\beta_U(\delta + it) = \left(\frac{t}{2\pi}\right)^{\delta/2-1/4} \omega_\delta(t) \Lambda_U^0(\delta + it; t) + O\left(t^{-1/4} U \log^3(2T)\right), \quad (27)$$

where $\omega_\delta(t)$ is smooth and

$$|\omega_\delta(t)| = 1 + O(T^{-1}), \quad \frac{d}{dt} |\omega_\delta(t)|^2 \ll T^{-2}.$$

Proof. At the single point $s = \sigma + it$, apply Lavrik's Theorem 1 [14, English pp. 130–131] to $\tilde{\Lambda}_{U,t}(s)$. Its parameter $\rho(t)$ has exactly the phase required in that theorem. This pointwise comparison is used only for the approximate functional equation; the fixed analytic function Λ_U remains the one used in the rectangle and Jensen arguments. The theorem's hypotheses hold because $Y_U(s) = \zeta(s) M_U(s) M_U(1-s)$ has the stated Riemann-type functional equation, its analytic half has only the inherited pole at $s = 1$, and (18) converges absolutely in the initial half-plane. The theorem gives

$$\begin{aligned} \tilde{\Lambda}_{U,t}(s) &= \sum_{\xi < \tau(t)} a_\xi \xi^{-s} + O\left(|M_U(0) M_U(1)| t^{\sigma/2-1/2}\right) \\ &\quad + O\left(t^{-\sigma/2} \sum_{\xi > 0} |a_\xi| e^{-\xi^2/t} \left(1 + \left|\sqrt{t} - \frac{\pi \xi^2}{\sqrt{t}}\right|\right)^{-1}\right). \end{aligned}$$

Lemma 5.2 changes the left side back to $\Lambda_U(s)$ at a cost $O(t^{-\sigma/2} U \log(2U))$, which is absorbed by the final error below. The first error is $O(t^{-\sigma/2} U \log(2U))$, since $|M_U(0)| \leq U$, $|M_U(1)| \ll \log(2U)$, and $\sigma \leq 1/2$. For the second, expand a_ξ using (18) and apply Lemma 5.1 with $A = \sqrt{t}$ and $h = \ell/m$:

$$\begin{aligned} &\sum_{\xi > 0} |a_\xi| e^{-\xi^2/t} \left(1 + \left|\sqrt{t} - \frac{\pi \xi^2}{\sqrt{t}}\right|\right)^{-1} \\ &\leq \sum_{\ell, m \leq U} \frac{1}{m} \sum_{n \geq 1} \frac{e^{-(n\ell/m)^2/t}}{1 + |\sqrt{t} - \pi(n\ell/m)^2/\sqrt{t}|} \\ &\ll \log(2T) \sum_{\ell, m \leq U} \left(\frac{1}{m} + \frac{1}{\ell}\right) \ll U \log(2U) \log(2T). \end{aligned}$$

This is stronger than the asserted logarithmic bound. Finally, Stirling's formula gives

$$\aleph(\delta + it) = \left(\frac{t}{2\pi}\right)^{\delta/2-1/4} \omega_\delta(t), \quad |\omega_\delta(t)| = 1 + O(T^{-1}), \quad \frac{d}{dt} |\omega_\delta(t)|^2 \ll T^{-2}.$$

The Stirling remainder multiplying the main polynomial is absorbed by the same elementary coefficient summation used above. This proves (27). $\square$

We use $\Delta_C \arg f$ for the net increment of a continuously chosen argument along an oriented curve C . It is not the total variation.

Lemma 5.4. *Let $U \asymp T^\theta$ be prime, where $0 < \theta < 1/4$ is fixed, and suppose*

$$b_1 = 1, \quad |b_n| \leq 1 \ (n \leq U), \quad b_U \neq 0.$$

Put

$$L = \log(2T), \quad \eta_U = \log^+ \frac{1}{|b_U|}.$$

Assume in addition that $\eta_U \ll L$, with an implied constant independent of T . This condition holds for every coefficient family used below. For fixed $0 \leq \sigma_1 < \sigma_2 \leq 3$ and $T \leq t \leq 2T$,

$$|\Delta_{[\sigma_1+it, \sigma_2+it]} \arg \Lambda_U| + |\Delta_{[\sigma_1+it, \sigma_2+it]} \arg \beta_U| \ll ZL\{L^2 + \eta_U + 1\}. \quad (28)$$

Proof. Use the paired analytic halves in (21). For $s = \sigma + it$, $\Lambda_U^-(\sigma - it) = \overline{\Lambda_U^+(\sigma + it)}$. Hence the zeros of the real and imaginary parts of $\Lambda_U^+(\sigma + it)$ on the horizontal segment are respectively the zeros there of

$$G_+(s) = \frac{\Lambda_U^+(s) + \Lambda_U^-(s - 2it)}{2}, \quad G_-(s) = \frac{\Lambda_U^+(s) - \Lambda_U^-(s - 2it)}{2i}.$$

Choose

$$\sigma_0 = AU\{L^2 + \eta_U + 1\} + \sigma_1 + \sigma_2, \quad s_0 = \sigma_0 + it,$$

where A is an absolute sufficiently large constant. Proposition A.1, proved from the explicit incomplete-gamma kernel in Appendix A, gives

$$\max_{|s-s_0| \leq 2\sigma_0} \log^+ |(s-1)(s-1-2it)G_\pm(s)| \ll ZL\{L^2 + \eta_U + 1\}.$$

The same proposition shows that at least one choice of sign satisfies

$$-\log |(s_0-1)(s_0-1-2it)G_\pm(s_0)| \ll ZL\{L^2 + \eta_U + 1\}.$$

The two linear factors remove the pole of $\Lambda_U^+(s)$ at $s = 1$ and that of $\Lambda_U^-(s - 2it)$ at $s = 1 + 2it$. Jensen's theorem between the concentric radii σ_0 and $2\sigma_0$ therefore bounds the number of zeros of the chosen $G_\pm$ in the inner disk by $O(ZL\{L^2 + \eta_U + 1\})$. The horizontal segment lies in that disk. Between two consecutive zeros of the chosen real or imaginary part, a continuous argument of Λ_U^+ changes by less than π . This proves the first term in (28). Finally, $\beta_U = \aleph \Lambda_U^+$, and uniform Stirling gives a horizontal argument increment $O(\log T)$ for $\aleph$; the second term follows. $\square$

For $0 < \alpha \leq 1$, define

$$S_b(\alpha, U) = \sum_{d \leq U^2} \Phi_\alpha(d) \left(\sum_{\substack{\ell, m \leq U \\ d | \ell m}} \frac{b_\ell b_m}{\ell m} \right)^2, \quad \Phi_\alpha(d) = d^\alpha \prod_{p|d} (1 - p^{-\alpha}).$$

Thus $S_b(1, U)$ is (16).

Lemma 5.5. *Let $1/4 \leq \delta < 1/2$, put $\alpha = 2\delta$ and $x = 1 - \alpha$. Uniformly for $X \geq 2$,*

$$\sum_{\xi \leq X} \frac{a_\xi^2}{\xi^\alpha} = \frac{X^x}{x} S_b(1, U) + \zeta(\alpha) Q_b(\alpha, U) + O(X^{-\alpha} R_b(U)), \quad (29)$$

where

$$Q_b(\alpha, U) = \sum_{d \leq U^2} \Phi_\alpha(d) \left(\sum_{\substack{\ell, m \leq U \\ d | \ell m}} \frac{b_\ell b_m}{\ell^\alpha m} \right)^2 \geq 0. \quad (30)$$

Since $\zeta(\alpha) < 0$ for $0 < \alpha < 1$, it follows that

$$\sum_{\xi \leq X} \frac{a_\xi^2}{\xi^{2\delta}} \leq \frac{X^x}{x} S_b(1, U) + O\left(X^{-2\delta} R_b(U)\right). \quad (31)$$

Proof. Expand the square using (18). The equality $n_1 \ell_1 / m_1 = n_2 \ell_2 / m_2$ is equivalent to $n_1 A = n_2 B$, where $A = \ell_1 m_2$ and $B = \ell_2 m_1$. Writing $g = (A, B)$, $A = ga$, $B = gb$, one has $n_1 = bk$, $n_2 = ak$, and the common frequency is kq , where $q = \ell_1 \ell_2 / g$. Euler summation, uniformly for $1/2 \leq \alpha < 1$ and $Y > 0$ (with an empty sum when $Y < 1$), gives

$$\sum_{k \leq Y} k^{-\alpha} = \frac{Y^{1-\alpha}}{1-\alpha} + \zeta(\alpha) + O(Y^{-\alpha}).$$

After multiplication by $q^{-\alpha}$ and summation over ℓ_i, m_i , the coefficient of X^x/x is

$$\sum_{\ell_1, m_1, \ell_2, m_2 \leq U} \frac{b_{\ell_1} b_{m_1} b_{\ell_2} b_{m_2}}{\ell_1 m_1 \ell_2 m_2} (\ell_1 m_2, \ell_2 m_1).$$

Interchanging m_1 and m_2 and using $\sum_{d|n} \varphi(d) = n$ identifies this with $S_b(1, U)$. The coefficient of $\zeta(\alpha)$ is

$$\sum_{\ell_1, m_1, \ell_2, m_2 \leq U} \frac{b_{\ell_1} b_{m_1} b_{\ell_2} b_{m_2} (\ell_1 m_2, \ell_2 m_1)^\alpha}{\ell_1^\alpha \ell_2^\alpha m_1 m_2}.$$

After the same interchange of m_1, m_2 and the identity $\sum_{d|n} \Phi_\alpha(d) = n^\alpha$, this is exactly $Q_b(\alpha, U)$ in (30). In particular it is non-negative. The Euler-summation remainder contributes $O(X^{-\alpha} R_b(U))$ after absolute values are taken. Finally, $\zeta(\alpha) < 0$ on $(0, 1)$, which proves (31). $\square$

Remark 5.6. The secondary quadratic form is necessarily the asymmetric form $Q_b(\alpha, U)$ in (30), not the symmetric expression obtained by formally replacing 1 by α in $S_b(1, U)$. This is forced by the factor $1/m$ in the generalised Dirichlet coefficient a_ξ . The translated source writes the secondary term in a symmetric notation; the direct expansion above removes that ambiguity. For the application only $Q_b(\alpha, U) \geq 0$ and $\zeta(\alpha) < 0$ are needed.

Lemma 5.7. *Let $\mathcal{F}(X, U)$ be the set of those frequencies $\xi < X$ that occur in (18), each counted once. Then*

$$\#\mathcal{F}(X, U) \ll XZ^2, \quad \min_{\substack{\xi, \eta \in \mathcal{F}(X, U) \\ \xi \neq \eta}} |\log \xi - \log \eta| \gg \frac{1}{XZ^2}.$$

Consequently, for arbitrary complex numbers c_ξ and arbitrary intervals $I_{\xi, \eta} \subseteq [T, 2T]$,

$$\left| \sum_{\xi \neq \eta} c_\xi \overline{c_\eta} \int_{I_{\xi, \eta}} t^{\delta-1/2} e^{it(\log \eta - \log \xi)} dt \right| \ll T^{\delta-1/2} XZ^2 \log(2XZ) \sum_{\xi} |c_\xi|^2. \quad (32)$$

Proof. Write a frequency in lowest terms as p/q . Since it arises as $n\ell/m$, its denominator divides m , so $q \leq U$; and $p < Xq$. This proves the cardinality bound. For two distinct reduced fractions $p/q, r/s$,

$$\left| \frac{p}{q} - \frac{r}{s} \right| \geq \frac{1}{qs} \geq \frac{1}{U^2}.$$

As both are at most X , $|\log(p/q) - \log(r/s)| \geq X^{-1}|p/q - r/s|$.

Order the logarithmic frequencies increasingly and denote their minimum spacing by Δ . Integration by parts gives

$$\left| \int_I t^{\delta-1/2} e^{it(u-v)} dt \right| \ll T^{\delta-1/2} |u-v|^{-1}.$$

For a fixed frequency, the k th neighbour on either side is at distance at least $k\Delta$. Hence every row sum of the absolute kernel is $O(T^{\delta-1/2} \Delta^{-1} \log(2\#\mathcal{F}))$. The Schur test (or $2|zw| \leq |z|^2 + |w|^2$) proves (32), since $\Delta^{-1} \ll XZ^2$. $\square$

Proposition 5.8. *Let the mollifier coefficients satisfy*

$$b_1 = 1, \quad |b_n| \leq 1 \ (n \leq U), \quad b_U \neq 0,$$

where $U \asymp T^\theta$ is prime, $0 < \theta < 1/4$, and

$$\log^+ \frac{1}{|b_U|} \ll \log(2T).$$

In particular,

$$U \log(2T) \left\{ \log^2(2T) + \log^+ \frac{1}{|b_U|} + 1 \right\} = o(T). \quad (33)$$

Put $x = 1 - 2\delta \asymp 1/\log T$. If

$$S_b(1, U) \leq \frac{\gamma + o(1)}{\log U}, \quad (34)$$

then

$$\frac{1}{T} \int_T^{2T} |\beta_U(\delta + it)|^2 dt \leq \frac{\gamma}{x \log U} + o(1). \quad (35)$$

The horizontal argument increments required in Lemma 4.1 are $o(T)$.

Proof. Write $X_1 = \sqrt{T/\pi}$, so every frequency occurring in the moving truncation $\xi < \tau(t)$ lies below X_1 . The diagonal contribution of the truncated polynomial is

$$D = \frac{1}{T} \int_T^{2T} \left(\frac{t}{2\pi} \right)^{\delta-1/2} \sum_{\xi < \sqrt{t/(2\pi)}} \frac{a_\xi^2}{\xi^{2\delta}} dt.$$

Since $x = 1 - 2\delta$ and $X = \sqrt{t/(2\pi)}$, Lemma 5.5 gives

$$\left(\frac{t}{2\pi} \right)^{\delta-1/2} \sum_{\xi < X} \frac{a_\xi^2}{\xi^{2\delta}} \leq \frac{S_b(1, U)}{x} + O\left(t^{-1/2} R_b(U)\right).$$

Consequently

$$D \leq \frac{S_b(1, U)}{x} + O(T^{-1/2} R_b(U)) = \frac{S_b(1, U)}{x} + o(1).$$

Moreover, (31), (34), and (23) show that

$$\sum_{\xi < X_1} \frac{|a_\xi|^2}{\xi^{2\delta}} \ll 1. \quad (36)$$

On expanding the square of the truncated polynomial, a pair $\xi \neq \eta$ is integrated over the interval

$$I_{\xi, \eta} = [\max(T, 2\pi\xi^2, 2\pi\eta^2), 2T],$$

with the interval understood to be empty if its left endpoint exceeds $2T$. The common factor $|\omega_\delta(t)|^2 = 1 + O(T^{-1})$ has derivative $O(T^{-2})$ and is absorbed in the same integration-by-parts estimate. Apply Lemma 5.7 with $c_\xi = a_\xi \xi^{-\delta}$ and $X = X_1$. After division by T , the complete off-diagonal is

$$\ll T^{-1} T^{\delta-1/2} X_1 U^2 \log(2TZ) \sum_{\xi < X_1} \frac{|a_\xi|^2}{\xi^{2\delta}} \ll T^{-1/2} U^2 \log(2T) = o(1).$$

This proves the mean square for the main truncated polynomial.

Lemma 5.3 gives a uniform remainder $E_U(t) \ll T^{-1/4} U \log^3 T = o(1)$. Using $|u + v|^2 \leq (1 + \rho)|u|^2 + (1 + \rho^{-1})|v|^2$ with $\rho \downarrow 0$ sufficiently slowly shows that the remainder and the cross term contribute $o(1)$ to the normalised mean square. This proves (35). Finally Lemma 5.4 and (33) give $o(T)$ for each horizontal boundary increment; the standard bounds for ζ and M_U on the auxiliary rectangle are smaller. $\square$

For the reciprocal-square-root coefficients,

$$|\lambda(n)| \leq 1, \quad \lambda(U) = -\frac{1}{2}$$

when U is prime. Proposition 5.8 therefore applies directly, and Theorem 9.6 supplies $\gamma = \mathcal{C}_{\text{res}}$.

Throughout the critical-line application we use the source normalisation (1). The mean-square proposition concerns the same unscaled analytic half β_U , so no compensating factor of $1/4$ occurs in (35).

6 The sharp reciprocal-square-root model

For $\text{Re } s > 1$, write

$$\zeta(s)^{-1/2} = \sum_{n \geq 1} \frac{\lambda(n)}{n^s}.$$

The function λ is multiplicative and

$$\lambda(p^k) = (-1)^k \binom{1/2}{k} \quad (k \geq 0),$$

so $\lambda(p) = -1/2$, $\lambda(p^2) = -1/8$, and so forth.

Squaring the Dirichlet series in its half-plane of absolute convergence gives

$$\left(\sum_{n \geq 1} \lambda(n) n^{-s} \right)^2 = \zeta(s)^{-1} = \sum_{n \geq 1} \mu(n) n^{-s},$$

so that, comparing Dirichlet coefficients,

$$(\lambda * \lambda)(n) = \mu(n) \quad (n \geq 1). \tag{37}$$

This exact convolution identity is what removes the same-side poles from the four-variable Euler product below.

Define the sharp truncated square

$$c_U(a) := \sum_{\substack{\ell m = a \\ \ell, m \leq U}} \lambda(\ell) \lambda(m), \quad a \leq U^2.$$

Then $c_U(a) = \mu(a)$ for $a \leq U$, but the boundary range $U < a \leq U^2$ is not Möbius: the two individual cutoffs remain visible.

The diagonal arithmetic sum attached to this model is

$$S_\lambda(U) := \sum_{d \leq U^2} \varphi(d) \left(\sum_{\substack{\ell, m \leq U \\ d | \ell m}} \frac{\lambda(\ell)\lambda(m)}{\ell m} \right)^2. \quad (38)$$

Lemma 6.1. *One has the exact identity*

$$S_\lambda(U) = \sum_{a, b \leq U^2} c_U(a)c_U(b) \frac{(a, b)}{ab}. \quad (39)$$

Proof. Expanding the square in (38) and grouping $a = \ell_1 m_1$ and $b = \ell_2 m_2$ gives

$$S_\lambda(U) = \sum_{a, b \leq U^2} \frac{c_U(a)c_U(b)}{ab} \sum_{d | (a, b)} \varphi(d).$$

The classical identity $\sum_{d | n} \varphi(d) = n$ yields (39). $\square$

These identities are exact. They do not yet give the asymptotic size of $S_\lambda(U)$.

7 The four-variable Euler product

For $\operatorname{Re} x_i > 0$, define

$$D(\mathbf{x}) = \sum_{\ell_1, m_1, \ell_2, m_2 \geq 1} \frac{\lambda(\ell_1)\lambda(m_1)\lambda(\ell_2)\lambda(m_2)(\ell_1 m_1, \ell_2 m_2)}{\ell_1^{1+x_1} m_1^{1+x_2} \ell_2^{1+x_3} m_2^{1+x_4}}. \quad (40)$$

It has the Euler product $D = \prod_p L_p$, where

$$L_p(\mathbf{x}) = \sum_{i, j, i', j' \geq 0} \lambda(p^i)\lambda(p^j)\lambda(p^{i'})\lambda(p^{j'}) p^{-\max(i+j, i'+j')} p^{-ix_1 - jx_2 - i'x_3 - j'x_4}. \quad (41)$$

Proposition 7.1. *There is $\eta > 0$ and an Euler product $\mathcal{A}(\mathbf{x})$, holomorphic for $|x_i| < \eta$ in a simply connected neighbourhood reached from $\operatorname{Re} x_i > 0$, such that*

$$D(\mathbf{x}) = \mathcal{A}(\mathbf{x}) \prod_{i=1}^4 \zeta(1+x_i)^{-1/2} \prod_{\substack{i \in \{1,2\} \\ j \in \{3,4\}}} \zeta(1+x_i+x_j)^{1/4}. \quad (42)$$

Moreover $\mathcal{A}(\mathbf{0}) = 1$.

Proof. At a fixed prime, terms of order p^{-1} in (41) arise in two ways. If exactly one of i, j, i', j' equals 1, the contribution is

$$-\frac{1}{2} \sum_{r=1}^4 p^{-1-x_r}.$$

If one exponent from $\{i, j\}$ and one from $\{i', j'\}$ equal 1, the total valuation on each side is 1, and the contribution is

$$\frac{1}{4} \sum_{\substack{r \in \{1,2\} \\ s \in \{3,4\}}} p^{-1-x_r-x_s}.$$

All other terms are $O(p^{-2+C\eta})$, uniformly for $|x_i| < \eta$. These are exactly the first-order terms of the local factors on the right of (42). Dividing L_p by those local factors therefore gives

$$\mathcal{A}_p(\mathbf{x}) = 1 + O(p^{-2+C\eta}),$$

so $\prod_p \mathcal{A}_p$ converges normally after reducing η if necessary.

At the origin, group the variables in (41) by $a = i + j$ and $b = i' + j'$. Identity (37) gives $\sum_{i+j=a} \lambda(p^i)\lambda(p^j) = \mu(p^a)$, hence

$$L_p(\mathbf{0}) = \sum_{a,b \in \{0,1\}} \mu(p^a)\mu(p^b)p^{-\max(a,b)} = 1 - \frac{1}{p}.$$

The product of the eight displayed singular local factors at the origin is also $(1-p^{-1})^2(1-p^{-1})^{-1} = 1 - p^{-1}$. Thus $\mathcal{A}_p(\mathbf{0}) = 1$ for every p , and $\mathcal{A}(\mathbf{0}) = 1$. $\square$

The same-side sums $x_1 + x_2$ and $x_3 + x_4$ do not occur in the polar part. At first order they would correspond to total valuation 2 on one side, and the identity $\mu(p^2) = 0$ removes them. The operative singularities are therefore the four bipartite quarter-poles.

For non-integral U , four applications of Perron's formula give, in the usual truncated-integral sense,

$$S_\lambda(U) = \frac{1}{(2\pi i)^4} \int_{(c_1)} \cdots \int_{(c_4)} D(\mathbf{x}) \frac{U^{x_1+x_2+x_3+x_4}}{x_1 x_2 x_3 x_4} d\mathbf{x}, \quad c_i > 0. \quad (43)$$

Formula (43) is exact after taking the truncation limits. The non-trivial missing step is to deform and localise these four contours with uniform error control.

8 The residue functional and its exact evaluation

The local scaling suggested by Proposition 7.1 is $x_i = u_i / \log U$. The four Perron kernels and the eight fractional zeta factors then have net homogeneity $(\log U)^{-1}$. Rather than treating a conditionally convergent four-dimensional inverse-Mellin (Bromwich) integral as a definition, we define the resulting constant by its non-negative inverse-Laplace form.

Let $\Omega \subset [0, \infty)^4$ consist of $\boldsymbol{\tau} = (\tau_{13}, \tau_{14}, \tau_{23}, \tau_{24})$ for which

$$\begin{aligned} c_1 &= 1 - \tau_{13} - \tau_{14} > 0, & c_2 &= 1 - \tau_{23} - \tau_{24} > 0, \\ c_3 &= 1 - \tau_{13} - \tau_{23} > 0, & c_4 &= 1 - \tau_{14} - \tau_{24} > 0. \end{aligned}$$

Define

$$\mathcal{C}_{\text{res}} := \frac{1}{\Gamma(\frac{1}{4})^4 \pi^2} \int_{\Omega} (\tau_{13}\tau_{14}\tau_{23}\tau_{24})^{-3/4} (c_1 c_2 c_3 c_4)^{-1/2} d\boldsymbol{\tau}. \quad (44)$$

The proof below both establishes convergence and evaluates the integral.

Theorem 8.1. *The integral (44) converges and*

$$\mathcal{C}_{\text{res}} = \frac{\Gamma(\frac{1}{4})^4}{2\pi^4} = \frac{2}{\Gamma(\frac{3}{4})^4} = 0.8869411685781154 \dots \quad (45)$$

Proof. Put

$$B = B\left(\frac{1}{4}, \frac{1}{2}\right) = \frac{\Gamma(\frac{1}{4})^2}{\sqrt{2\pi}}, \quad F(t) = {}_2F_1\left(\frac{1}{2}, \frac{1}{4}; \frac{3}{4}; t\right).$$

Set $y = 1 - \tau_{14}$ and $z = 1 - \tau_{23}$. For fixed $y, z \in (0, 1)$, the τ_{13} and τ_{24} integrations separate and each equals

$$g(y, z) = \int_0^{\min(y, z)} t^{-3/4} \{(y-t)(z-t)\}^{-1/2} dt = m^{-1/4} M^{-1/2} B F(m/M),$$

where $m = \min(y, z)$ and $M = \max(y, z)$. This follows from $t = ms$ and Euler's integral for ${}_2F_1$. Tonelli's theorem applies because every integrand is non-negative. By symmetry in y, z ,

$$\begin{aligned}\Gamma(\tfrac{1}{4})^4 \pi^2 \mathcal{C}_{\text{res}} &= 2B^2 \int_0^1 (1-y)^{-3/4} y^{-1} \int_0^y (1-z)^{-3/4} z^{-1/2} F(z/y)^2 dz dy \\ &= 2B^2 \int_0^1 t^{-1/2} F(t)^2 Y(t) dt,\end{aligned}\tag{46}$$

where, after $z = yt$,

$$\begin{aligned}Y(t) &= \int_0^1 y^{-1/2} (1-y)^{-3/4} (1-yt)^{-3/4} dy \\ &= B \left(\frac{1}{2}, \frac{1}{4} \right) {}_2F_1 \left(\frac{3}{4}, \frac{1}{2}; \frac{3}{4}; t \right) = B(1-t)^{-1/2}.\end{aligned}$$

It remains to use the following identity, proved in Lemma 8.2:

$$\int_0^1 t^{-1/2} (1-t)^{-1/2} F(t)^2 dt = B.\tag{47}$$

Equations (46) and (47) give

$$\Gamma(\tfrac{1}{4})^4 \pi^2 \mathcal{C}_{\text{res}} = 2B^4.$$

Since $B = \Gamma(\frac{1}{4})^2 / \sqrt{2\pi}$, this is $\mathcal{C}_{\text{res}} = \Gamma(\frac{1}{4})^4 / (2\pi^4)$. The second form follows from $\Gamma(\frac{1}{4})\Gamma(\frac{3}{4}) = \pi\sqrt{2}$. $\square$

Lemma 8.2. For $F(t) = {}_2F_1(\frac{1}{2}, \frac{1}{4}; \frac{3}{4}; t)$,

$$\int_0^1 t^{-1/2} (1-t)^{-1/2} F(t)^2 dt = B \left(\frac{1}{4}, \frac{1}{2} \right).$$

Proof. All integrands below are non-negative. Consequently Tonelli's theorem justifies the interchanges, initially with values in $[0, \infty]$; the final finite value then proves convergence throughout. Write $B = B(\frac{1}{4}, \frac{1}{2})$. Euler's integral gives

$$F(t) = B^{-1} \int_0^1 s^{-3/4} (1-s)^{-1/2} (1-ts)^{-1/2} ds.$$

Opening both factors of F yields

$$B^2 K = \int_0^1 \int_0^1 (s\sigma)^{-3/4} [(1-s)(1-\sigma)]^{-1/2} W(s, \sigma) ds d\sigma,$$

where

$$W(s, \sigma) = \int_0^1 \frac{t^{-1/2} (1-t)^{-1/2}}{\sqrt{(1-st)(1-\sigma t)}} dt.$$

The substitution $t = u/(1-\sigma + \sigma u)$ gives

$$W(s, \sigma) = \pi(1-\sigma)^{-1/2} {}_2F_1 \left(\frac{1}{2}, \frac{1}{2}; 1; \frac{s-\sigma}{1-\sigma} \right).$$

Use symmetry to restrict to $0 < \sigma < s < 1$ and double. With $\xi = (s-\sigma)/(1-\sigma)$, equivalently $\sigma = (s-\xi)/(1-\xi)$, one obtains

$$B^2 K = 2\pi \int_0^1 s^{-3/4} (1-s)^{-1/2} \int_0^s (s-\xi)^{-3/4} (1-\xi)^{-1/4} {}_2F_1 \left(\frac{1}{2}, \frac{1}{2}; 1; \xi \right) d\xi ds.$$

Interchange the integrations. The substitution $s = \xi + (1 - \xi)r$ and Euler's integral show that

$$\int_{\xi}^1 s^{-3/4}(1-s)^{-1/2}(s-\xi)^{-3/4} ds = B \xi^{-1/2}(1-\xi)^{-1/4}.$$

Therefore

$$B^2 K = 2\pi B \int_0^1 \xi^{-1/2}(1-\xi)^{-1/2} {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; \xi\right) d\xi.$$

Termwise integration, justified by non-negativity, gives

$$\begin{aligned} \int_0^1 \xi^{-1/2}(1-\xi)^{-1/2} {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; \xi\right) d\xi &= \pi {}_3F_2\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}; 1, 1; 1\right) \\ &= \frac{\pi^2}{\Gamma(\frac{3}{4})^4}. \end{aligned}$$

For the last equality, Clausen's identity gives

$${}_3F_2\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}; 1, 1; z\right) = {}_2F_1\left(\frac{1}{4}, \frac{1}{4}; 1; z\right)^2,$$

and Gauss's theorem at $z = 1$ gives ${}_2F_1(\frac{1}{4}, \frac{1}{4}; 1; 1) = \sqrt{\pi}/\Gamma(\frac{3}{4})^2$. Thus

$$B^2 K = \frac{2\pi^3 B}{\Gamma(\frac{3}{4})^4} = B^3,$$

where the final equality is the reflection formula $\Gamma(\frac{1}{4})\Gamma(\frac{3}{4}) = \pi\sqrt{2}$. Hence $K = B$. $\square$

Remark 8.3. Theorem 8.1 evaluates the local residue functional forced by the polar factorisation. Theorem 9.6 identifies this functional with the sharp-cutoff limit $\lim_{U \rightarrow \infty} (\log U) S_{\lambda}(U)$.

9 The sharp-box transfer theorem

The sharp-box transfer is proved by separating the signed part of the Euler product into four one-variable Selberg–Delange tails. The four cross factors have non-negative coefficients, which permits a direct logarithmic-measure argument.

For $z \in \mathbb{C}$, let $\tau_z(n)$ be defined by

$$\zeta(s)^z = \sum_{n \geq 1} \frac{\tau_z(n)}{n^s} \quad (\operatorname{Re} s > 1),$$

with the branch that is positive for real $s > 1$. Thus $\tau_{-1/2} = \lambda$ and $\tau_{1/4}(n) \geq 0$. Put

$$\alpha = \frac{1}{4}, \quad \beta = \frac{1}{2}, \quad L(X) = \sum_{n \leq X} \frac{\lambda(n)}{n},$$

where $L(X) = 0$ for $0 < X < 1$.

Lemma 9.1. *As $t \rightarrow \infty$,*

$$\sum_{n \leq e^t} \frac{\tau_{1/4}(n)}{n} = \frac{t^{1/4}}{\Gamma(5/4)} + O(t^{-3/4}), \quad (48)$$

$$L(e^t) = \frac{t^{-1/2}}{\Gamma(1/2)} + O(t^{-3/2}). \quad (49)$$

Moreover, for $t \geq 0$,

$$|L(e^t)| \ll (1+t)^{-1/2}. \quad (50)$$

Proof. This is the classical Selberg–Delange theorem applied, after Perron inversion, to $\zeta(1+s)^z$ with $z = 1/4$ and $z = -1/2$. For completeness, near $s = 0$ one has

$$\zeta(1+s)^z = s^{-z}\{1 + O_z(s)\}.$$

After moving the Perron contour to a Hankel contour around the negative real axis in the standard zero-free region, the leading inverse-Laplace integral is

$$\frac{1}{2\pi i} \int_{\mathcal{H}} e^{ts} s^{-z-1} ds = \frac{t^z}{\Gamma(z+1)}.$$

The next term in the local expansion is $O(t^{\operatorname{Re} z - 1})$, while the remaining contour is smaller than every fixed negative power of t . This gives (48) and (49). Equation (50) follows from (49) for $t \geq 2$ and by enlarging the implied constant on the compact interval $0 \leq t \leq 2$. More precisely, apply [8, Theorem 1.1, pp. 350–351] with $f = \tau_z$, $\varrho = z$, and regular factor $G = 1$, first for $z = 1/4$ and then for $z = -1/2$, and use partial summation. The branch in that theorem is the one fixed above; its arbitrary parameter A supplies the two displayed remainders. $\square$

Let $R \geq 2$ and introduce the positive measure

$$d\nu_R(u) = R^{-1/4} \sum_{n \geq 1} \frac{\tau_{1/4}(n)}{n} \delta_{\log n/R}(u), \quad (51)$$

and the rescaled signed tail

$$W_R(v) = R^{1/2} L(e^{Rv}), \quad v \geq 0. \quad (52)$$

Here and below, $\mu_R \Rightarrow \mu$ denotes weak convergence of measures, that is, $\int f d\mu_R \rightarrow \int f d\mu$ for every bounded continuous f supported in the interval concerned.

Lemma 9.2. *On every fixed compact interval of $[0, \infty)$,*

$$\nu_R \Rightarrow \frac{u^{-3/4}}{\Gamma(1/4)} du.$$

Also, locally uniformly for $v > 0$,

$$W_R(v) \longrightarrow \frac{v^{-1/2}}{\sqrt{\pi}}, \quad |W_R(v)| \ll (v + R^{-1})^{-1/2}.$$

Finally, if $I_k = [k/R, (k+1)/R)$, then uniformly for $k \geq 0$,

$$\nu_R(I_k) \ll R^{-1} \left(\frac{k+1}{R} \right)^{-3/4}. \quad (53)$$

Proof. The convergence of distribution functions follows from (48): for fixed $u > 0$,

$$\nu_R([0, u]) = R^{-1/4} \sum_{n \leq e^{Ru}} \frac{\tau_{1/4}(n)}{n} = \frac{u^{1/4}}{\Gamma(5/4)} + O(R^{-1}u^{-3/4}).$$

The assertions about W_R follow from (49) and (50). For $k \geq 1$, subtracting (48) at $t = k+1$ and $t = k$ gives

$$\sum_{e^k < n \leq e^{k+1}} \frac{\tau_{1/4}(n)}{n} \ll k^{-3/4}.$$

After multiplication by $R^{-1/4}$ this is (53); the case $k = 0$ is immediate. $\square$

Let

$$E = \{13, 14, 23, 24\},$$

and, for $\mathbf{q} = (q_e)_{e \in E}$, put

$$Q_1 = q_{13}q_{14}, \quad Q_2 = q_{23}q_{24}, \quad Q_3 = q_{13}q_{23}, \quad Q_4 = q_{14}q_{24}.$$

Define the polar model

$$D_0(\mathbf{x}) = \prod_{i=1}^4 \zeta(1+x_i)^{-1/2} \prod_{\substack{i \in \{1,2\} \\ j \in \{3,4\}}} \zeta(1+x_i+x_j)^{1/4}. \quad (54)$$

If $S_0(Y_1, Y_2, Y_3, Y_4)$ denotes the sharp box sum of the Dirichlet coefficients of D_0 , then expanding first the four cross factors gives the exact identity

$$S_0(\mathbf{Y}) = \sum_{\substack{q_e \geq 1 \\ Q_i \leq Y_i \ (1 \leq i \leq 4)}} \prod_{e \in E} \frac{\tau_{1/4}(q_e)}{q_e} \prod_{i=1}^4 L(Y_i/Q_i). \quad (55)$$

For $\mathbf{b} = (b_1, b_2, b_3, b_4)$ with positive coordinates, let $\mathcal{P}(\mathbf{b})$ be the polytope of $\mathbf{u} = (u_e)_{e \in E} \geq 0$ for which

$$\begin{aligned} c_1 &= b_1 - u_{13} - u_{14} \geq 0, & c_2 &= b_2 - u_{23} - u_{24} \geq 0, \\ c_3 &= b_3 - u_{13} - u_{23} \geq 0, & c_4 &= b_4 - u_{14} - u_{24} \geq 0. \end{aligned}$$

Set

$$\mathfrak{C}(\mathbf{b}) = \frac{1}{\Gamma(1/4)^4 \pi^2} \int_{\mathcal{P}(\mathbf{b})} \prod_{e \in E} u_e^{-3/4} \prod_{i=1}^4 c_i^{-1/2} d\mathbf{u}. \quad (56)$$

Lemma 9.3. *The integral in (56) is locally finite. Moreover, on every compact neighbourhood of $\mathbf{1}$ its integrand has a common integrable majorant after the natural translation of the moving faces; consequently $\mathfrak{C}(\mathbf{b})$ is continuous there.*

Proof. It is enough first to treat $\mathbf{b} = \mathbf{1}$. Put

$$g(y, z) = \int_0^{\min(y, z)} t^{-3/4} (y-t)^{-1/2} (z-t)^{-1/2} dt.$$

Integrating successively in u_{13} and u_{24} reduces the four-dimensional integral, up to its fixed normalising factor, to

$$\int_0^1 \int_0^1 u^{-3/4} v^{-3/4} g(1-u, 1-v)^2 du dv.$$

If $m = \min(y, z)$ and $M = \max(y, z)$, Euler's integral gives

$$g(y, z) = B(\tfrac{1}{4}, \tfrac{1}{2}) m^{-1/4} M^{-1/2} {}_2F_1(\tfrac{1}{2}, \tfrac{1}{4}; \tfrac{3}{4}; m/M).$$

The hypergeometric function is zero-balanced, and hence

$$g(y, z) \ll m^{-1/4} M^{-1/2} \left(1 + \log^+ \frac{M}{|M-m|} \right).$$

Consider, by symmetry, the region $0 < y \leq z < 1$. Near $y = z = 0$ the resulting majorant is

$$y^{-1/2} z^{-1} \left(1 + \log \frac{z}{z-y} \right)^2,$$

whose y -integral is $O(z^{1/2})$ after the substitution $y = zr$; the remaining $z^{-1/2}$ singularity is integrable. Near $y = z = 1$, the only power singularities are $(1 - y)^{-3/4}(1 - z)^{-3/4}$, and the logarithmic singularity along $y = z$ is locally integrable. The mixed boundary regions are easier. This proves local finiteness.

If $\mathbf{b}$ ranges in a sufficiently small compact neighbourhood of $\mathbf{1}$, translate each moving face by the corresponding change in b_i . The preceding estimates remain valid with uniform constants and on a fixed slightly enlarged compact domain. They therefore provide a common integrable majorant. Dominated convergence proves the continuity of $\mathfrak{C}(\mathbf{b})$. $\square$

In particular,

$$\mathfrak{C}(1, 1, 1, 1) = \mathcal{C}_{\text{res}}. \quad (57)$$

Proposition 9.4. *Uniformly for $\mathbf{b}$ in a sufficiently small fixed compact neighbourhood of $(1, 1, 1, 1)$,*

$$S_0(e^{Rb_1}, e^{Rb_2}, e^{Rb_3}, e^{Rb_4}) = \frac{\mathfrak{C}(\mathbf{b})}{R} + o(R^{-1}). \quad (58)$$

Proof. From (55), (51), and (52), one obtains the exact rescaled identity

$$RS_0(e^{Rb_1}, e^{Rb_2}, e^{Rb_3}, e^{Rb_4}) = \int_{\mathcal{P}(\mathbf{b})} \prod_{i=1}^4 W_R(c_i) d\nu_R^{\otimes 4}(\mathbf{u}). \quad (59)$$

We justify passage to the limit, including all boundary faces.

Fix $\varepsilon > 0$ and first restrict to the compact subset on which $u_e \geq \varepsilon$ for every e and $c_i \geq \varepsilon$ for every i . Lemma 9.2, weak convergence of product measures, and local uniform convergence of W_R give the required limit there, uniformly for $\mathbf{b}$ in the stated compact set.

It remains to prove uniform integrability. Partition each u_e -axis into the intervals I_k of Lemma 9.2. By (53), the $\nu_R^{\otimes 4}$ -mass of a mesh box $B = \prod_e I_{k_e}$ is at most a constant times

$$R^{-4} \prod_{e \in E} (u_{e,B} + R^{-1})^{-3/4},$$

where $u_{e,B} = k_e/R$. On a box meeting $\mathcal{P}(\mathbf{b})$, Lemma 9.2 bounds the kernel by a constant times

$$\prod_{i=1}^4 (c_{i,B}^- + R^{-1})^{-1/2}, \quad c_{i,B}^- = \max\left(0, b_i - \sum_{e \ni i} \frac{k_e + 1}{R}\right).$$

Consequently the contribution of any collection $\mathcal{B}$ of such mesh boxes is bounded by a constant times the corresponding singular Riemann sum. To compare this sum with an integral, take in every mesh box a fixed sub-box of side $1/(8R)$ adjacent to its upper corner. On that sub-box all coordinate factors and all regularised deficit factors are comparable, with constants independent of the box, R , and $\mathbf{b}$. The sub-boxes are disjoint. It follows that

$$\begin{aligned} & \int_{\bigcup \mathcal{B} \cap \mathcal{P}(\mathbf{b})} \prod_i |W_R(c_i)| d\nu_R^{\otimes 4} \\ & \ll \int_{\mathcal{N}_{3/R}(\bigcup \mathcal{B}) \cap \mathcal{P}(\mathbf{b} + 3R^{-1}\mathbf{1})} \prod_{e \in E} u_e^{-3/4} \prod_{i=1}^4 \left(b_i + 3R^{-1} - \sum_{e \ni i} u_e\right)^{-1/2} d\mathbf{u}. \end{aligned} \quad (60)$$

Here $\mathcal{N}_{3/R}$ denotes a $3/R$ -neighbourhood. The enlargement by $3/R$ merely ensures that every regularised mesh factor is dominated by the corresponding unregularised factor on the enlarged polytope.

The right-hand side of (60) is uniformly bounded by Lemma 9.3. More importantly, if $\mathcal{B}$ consists of boxes meeting one of the sets $u_e < \varepsilon$ or $c_i < \varepsilon$, then the right-hand side tends to zero

uniformly as $\varepsilon \downarrow 0$, after first taking $R \rightarrow \infty$. This is simply the absolute continuity of the finite integrals (56); the $3/R$ enlargement disappears in the limit. Thus the family of integrands in (59) is uniformly integrable.

Combining the interior convergence with this boundary estimate gives

$$RS_0(e^{R\mathbf{b}}) \longrightarrow \mathfrak{C}(\mathbf{b})$$

uniformly in the stated neighbourhood, proving (58). $\square$

We next restore the regular Euler product $\mathcal{A}$ from Proposition 7.1.

Lemma 9.5. *There is $\eta > 0$ and coefficients $a(\mathbf{r})$, indexed by $\mathbf{r} = (r_1, r_2, r_3, r_4) \in \mathbb{N}^4$, such that*

$$\mathcal{A}(\mathbf{x}) = \sum_{\mathbf{r} \in \mathbb{N}^4} \frac{a(\mathbf{r})}{r_1^{x_1} r_2^{x_2} r_3^{x_3} r_4^{x_4}}, \quad \sum_{\mathbf{r}} |a(\mathbf{r})| (r_1 r_2 r_3 r_4)^\eta < \infty. \quad (61)$$

Moreover,

$$\sum_{\mathbf{r}} a(\mathbf{r}) = \mathcal{A}(\mathbf{0}) = 1. \quad (62)$$

Proof. The proof of Proposition 7.1 gives, after the first-order local terms have been divided out,

$$\mathcal{A}_p(\mathbf{x}) = 1 + O(p^{-2+C\eta})$$

uniformly for $|\operatorname{Re} x_i| \leq \eta$, once η is small enough. Expanding the local binomial factors and the absolutely convergent local sum (41) shows more precisely that the sum of the absolute values of the non-constant local Dirichlet coefficients, weighted by $p^{\eta(\nu_1 + \dots + \nu_4)}$, is $O(p^{-2+C\eta})$. The product over p therefore converges in the weighted ℓ^1 coefficient norm. This gives (61). Absolute convergence at $\mathbf{x} = \mathbf{0}$ and Proposition 7.1 give (62). $\square$

Theorem 9.6. *As $U \rightarrow \infty$,*

$$S_\lambda(U) = \frac{c_{\text{res}}}{\log U} + o\left(\frac{1}{\log U}\right). \quad (63)$$

Proof. Write $R = \log U$. Let $b_0(\mathbf{n})$ denote the Dirichlet coefficients of D_0 in the variables $\mathbf{x}$, so that $S_0(\mathbf{Y})$ is their box sum. Equations (42) and (61) imply the exact coordinatewise convolution identity

$$S_\lambda(U) = \sum_{\mathbf{r}} a(\mathbf{r}) S_0(U/r_1, U/r_2, U/r_3, U/r_4), \quad (64)$$

with the convention that $S_0(\mathbf{Y}) = 0$ if some $Y_i < 1$.

We first record a crude bound. The Euler products at $1 + 1/R$ give

$$\sum_{n \leq U} \frac{|\lambda(n)|}{n} \ll R^{1/2}, \quad \sum_{n \leq U} \frac{\tau_{1/4}(n)}{n} \ll R^{1/4}.$$

Indeed, the first absolute-value Dirichlet series has first prime coefficient $1/2$ and hence is $\ll \zeta(1 + 1/R)^{1/2}$, while the second is exactly $\zeta(1 + 1/R)^{1/4}$; Rankin's trick removes the sharp cutoff. From (55), uniformly for $1 \leq Y_i \leq U$,

$$|S_0(\mathbf{Y})| \ll R^3. \quad (65)$$

Split (64) at $\max_i r_i \leq e^{\sqrt{R}}$. By (61) and (65), the tail is

$$\ll R^3 e^{-\eta\sqrt{R}} \sum_{\mathbf{r}} |a(\mathbf{r})| (r_1 r_2 r_3 r_4)^\eta = o(R^{-1}).$$

In the remaining range put

$$b_i = 1 - \frac{\log r_i}{R}.$$

Then $1 - R^{-1/2} \leq b_i \leq 1$. Proposition 9.4, uniformly in this range, and the continuity of $\mathfrak{C}$ at $\mathbf{1}$ give

$$RS_0(U/r_1, U/r_2, U/r_3, U/r_4) = \mathcal{C}_{\text{res}} + o(1),$$

where the $o(1)$ is uniform in $\mathbf{r}$. Therefore

$$\begin{aligned} RS_\lambda(U) &= \{\mathcal{C}_{\text{res}} + o(1)\} \sum_{\max r_i \leq e^{\sqrt{R}}} a(\mathbf{r}) + o(1) \\ &= \mathcal{C}_{\text{res}} + o(1) \end{aligned}$$

by (62) and weighted absolute convergence. This is (63). $\square$

Remark 9.7. The proof does not require a multivariable Tauberian theorem for a signed arithmetic function. The four signed factors are treated separately by the one-variable estimate (49); after that separation, the four edge measures in (51) are positive. This is why the non-negativity restriction in the multivariable theorem of de la Bretèche [7] is avoided.

10 Weighted power transfer

For $a > 0$ put

$$b_{n,a}(U) = \lambda(n) \left(\frac{\log(U/n)}{\log U} \right)^a \quad (n \leq U)$$

and let $S_a(U)$ denote (16) for these coefficients. Write $R = \log U$ and

$$L_{a,R}(c) = \sum_{n \leq e^{Rc}} \frac{\lambda(n)}{n} \left(c - \frac{\log n}{R} \right)^a, \quad W_{a,R}(c) = R^{1/2} L_{a,R}(c).$$

If $L(X) = \sum_{n \leq X} \lambda(n)/n$, then Stieltjes summation gives

$$L_{a,R}(c) = a \int_0^c L(e^{R(c-v)}) v^{a-1} dv. \quad (66)$$

The one-variable estimate (49) and (50) imply, locally uniformly for $c > 0$,

$$W_{a,R}(c) \longrightarrow K_a c^{a-1/2}, \quad K_a = \frac{\Gamma(a+1)}{\Gamma(a+1/2)}, \quad (67)$$

and

$$|W_{a,R}(c)| \ll_a (c + R^{-1})^{a-1/2}. \quad (68)$$

On the polytope Ω of Section 8, define

$$\mathcal{C}_a = \frac{K_a^4}{\Gamma(1/4)^4} \int_{\Omega} (\tau_{13}\tau_{14}\tau_{23}\tau_{24})^{-3/4} (c_1 c_2 c_3 c_4)^{a-1/2} d\boldsymbol{\tau}. \quad (69)$$

Lemma 10.1. *Let $\mathbf{b} = (b_1, b_2, b_3, b_4)$ lie in a fixed neighbourhood of $\mathbf{1}$, and impose the coordinate caps $n_i \leq e^{Rb_i}$ in the polar model D_0 . Weight the i th coordinate by*

$$\left(b_i - \frac{\log n_i}{R} \right)^a.$$

Denote the resulting polar box sum by $S_{a,0,R}(\mathbf{b})$. Then, uniformly for $\mathbf{b} \rightarrow \mathbf{1}$,

$$RS_{a,0,R}(\mathbf{b}) \longrightarrow \mathfrak{C}_a(\mathbf{b}), \quad (70)$$

where

$$\mathfrak{C}_a(\mathbf{b}) = \frac{K_a^4}{\Gamma(1/4)^4} \int_{\mathcal{P}(\mathbf{b})} \prod_{e \in E} u_e^{-3/4} \prod_{i=1}^4 \left(b_i - \sum_{e \ni i} u_e \right)^{a-1/2} du.$$

In particular $\mathfrak{C}_a(\mathbf{1}) = \mathcal{C}_a$.

Proof. After expansion of the four positive cross factors, the edge variables are again governed by the positive measures ν_R from Proposition 9.4; the four coordinate tails are $W_{a,R}(c_i)$. On every compact subset of the interior, (67) gives uniform convergence to $K_a c_i^{a-1/2}$. Equation (68) gives the mesh majorant

$$\prod_{e \in E} (u_e + R^{-1})^{-3/4} \prod_{i=1}^4 (c_i + R^{-1})^{a-1/2}.$$

If $0 < a < 1/2$, this is bounded by a constant times the sharp majorant with $c_i^{-1/2}$; if $a \geq 1/2$, the deficit factors are locally bounded. The sub-box comparison used in (60) therefore applies uniformly for $\mathbf{b}$ near $\mathbf{1}$. Boundary boxes meeting $u_e < \varepsilon$ or $c_i < \varepsilon$ contribute $o_\varepsilon(1)$ uniformly after $R \rightarrow \infty$, while the interior boxes converge as ordinary Riemann sums. Letting $\varepsilon \downarrow 0$ proves (70). Local dominated convergence also shows that $\mathfrak{C}_a(\mathbf{b})$ is continuous at $\mathbf{1}$. $\square$

Lemma 10.2. *Let $a(\mathbf{r})$ be the coefficients in Lemma 9.5. Then*

$$S_a(e^R) = \sum_{\mathbf{r}} a(\mathbf{r}) S_{a,0,R} \left(1 - \frac{\log r_1}{R}, \dots, 1 - \frac{\log r_4}{R} \right), \quad (71)$$

with the convention that a term is zero if some $r_i > e^R$. Moreover,

$$RS_a(e^R) = \mathcal{C}_a + o(1). \quad (72)$$

Proof. The Dirichlet-series identity $D = \mathcal{A}D_0$ gives coordinatewise convolution. If the regular factor contributes r_i to the i th coordinate, then

$$\left(1 - \frac{\log(r_i n_i)}{R} \right)^a = \left(b_i - \frac{\log n_i}{R} \right)^a, \quad b_i = 1 - \frac{\log r_i}{R},$$

which proves the exact identity (71).

Since the weights lie in $[0, 1]$, the absolute polar sum satisfies the same crude estimate as (65):

$$|S_{a,0,R}(\mathbf{b})| \ll R^3 \quad (0 \leq b_i \leq 1). \quad (73)$$

Split (71) at $\max_i r_i \leq e^{\sqrt{R}}$. Weighted absolute convergence from (61) and (73) give for the tail

$$\ll R^3 e^{-\eta\sqrt{R}} \sum_{\mathbf{r}} |a(\mathbf{r})| (r_1 r_2 r_3 r_4)^\eta = o(R^{-1}).$$

In the remaining range, $1 - R^{-1/2} \leq b_i \leq 1$. Lemma 10.1 and continuity at $\mathbf{1}$ give, uniformly in $\mathbf{r}$,

$$RS_{a,0,R}(\mathbf{b}) = \mathcal{C}_a + o(1).$$

Therefore

$$RS_a(e^R) = \{\mathcal{C}_a + o(1)\} \sum_{\max r_i \leq e^{\sqrt{R}}} a(\mathbf{r}) + o(1) = \mathcal{C}_a + o(1),$$

using (62). This proves (72). $\square$

Theorem 10.3. *For every fixed $a > 0$,*

$$S_a(U) = \frac{\mathcal{C}_a}{\log U} + o\left(\frac{1}{\log U}\right).$$

Proof. Take $R = \log U$ in Lemma 10.2. □

10.1 Zhuravlev's coefficients evaluated exactly

The transfer theorems above use λ only through Proposition 7.1 and through $\sum_{\mathbf{r}} a(\mathbf{r}) = \mathcal{A}(\mathbf{0})$. Both survive for a general squarefree-supported multiplicative b with $b(p) = -v_p$ and $v_p \rightarrow \frac{1}{2}$, because the first-order terms of the local factor are unchanged: exactly one exponent equal to 1 contributes $-v_p \sum_r p^{-1-x_r}$, one exponent from each side contributes $v_p^2 \sum_{r,s} p^{-1-x_r-x_s}$, and everything else is $O(p^{-2})$, the same-side terms now having total valuation two rather than vanishing coefficient. Writing $w_0 = 1$, $w_1 = -2v_p$, $w_2 = v_p^2$ for the inner sums $\sum_{i+j=k} b(p^i)b(p^j)$ and using $L_p(\mathbf{0}) = \sum_{k,l \leq 2} w_k w_l p^{-\max(k,l)}$, one gets

$$\mathcal{A}_b(\mathbf{0}) = \prod_p \frac{L_p(\mathbf{0})}{1 - p^{-1}}, \quad L_p(\mathbf{0}) = 1 + \frac{4v_p(v_p - 1)}{p} + \frac{v_p^2(2 - 4v_p + v_p^2)}{p^2}, \quad (74)$$

and, for coefficients carrying the profile of exponent a ,

$$S_b(1, U) = \frac{\mathcal{A}_b(\mathbf{0}) \mathcal{C}_a}{\log U} + o\left(\frac{1}{\log U}\right). \quad (75)$$

Three consequences bear on Section 3.2.

The normalisation $pf(p) \rightarrow -\frac{1}{2}$ is forced, not chosen. For constant v the polar exponents in (42) become $-v$ and v^2 , the net homogeneity of the Perron integral is $(\log U)^{4v(v-1)}$, and

$$4v(v-1) = 4\left(v - \frac{1}{2}\right)^2 - 1 \geq -1,$$

with equality only at $v = \frac{1}{2}$. Any other limiting value costs a power of $\log U$, not merely a constant. Zhuravlev's $f(p) \sim -1/(2p)$ is therefore the unique admissible normalisation, and it is exactly $\lambda(p) = -\frac{1}{2}$ recovered on squarefree support.

Squarefree support is cheap. For $b = \mu^2 \lambda$ one has $v_p \equiv \frac{1}{2}$, so $L_p(\mathbf{0}) = 1 - p^{-1} + \frac{1}{16}p^{-2}$ and

$$\mathcal{A}_{\mu^2 \lambda}(\mathbf{0}) = \prod_p \left(1 + \frac{1}{16p(p-1)}\right) = 1.0489438\dots, \quad (76)$$

so discarding the prime powers of λ costs under 5%.

Zhuravlev's normalisation supplies a profile. Since $b_\ell = \ell f(\ell) N(U/\ell; \ell)/N(U)$ with $N(x; \ell) = \sum_{q \leq x, (q, \ell)=1} |f(q)|$, and since $|f|$ has the Dirichlet series $\prod_p (1 + |f(p)|p^{-s}) = \zeta(1+s)^{1/2} \times (\text{regular})$, one has $N(x) \asymp (\log x)^{1/2}$ and hence

$$\frac{N(U/\ell)}{N(U)} \sim \left(\frac{\log(U/\ell)}{\log U}\right)^{1/2}.$$

His coefficients are thus, up to a convergent Euler product, the power profile of Section 10 with $a = \frac{1}{2}$. At that exponent the factors $c_i^{a-1/2}$ in (69) disappear and $\mathcal{C}_{1/2}$ is elementary: for fixed τ_{14}, τ_{23} the remaining two variables are each free up to $m = 1 - \max(\tau_{14}, \tau_{23})$, so the inner integrals factor as $(4m^{1/4})^2$ and

$$\int_{\Omega} (\tau_{13} \tau_{14} \tau_{23} \tau_{24})^{-3/4} d\boldsymbol{\tau} = 128 \int_0^1 y^{-1/2} (1-y)^{1/2} dy = 128 B\left(\frac{1}{2}, \frac{3}{2}\right) = 64\pi.$$

With $K_{1/2} = \sqrt{\pi}/2$ this gives the closed form

$$\mathcal{C}_{1/2} = \frac{4\pi^3}{\Gamma(\frac{1}{4})^4} = \frac{\Gamma(\frac{3}{4})^4}{\pi} = 0.7177700110\dots, \quad (77)$$

against $\mathcal{C}_0 = \mathcal{C}_{\text{res}} = 0.8869411\dots$; the profile is worth a factor $\Gamma(\frac{3}{4})^8/(2\pi) = 0.8093$. As a check, (77) also follows from the one-dimensional reduction (79) at $a = \frac{1}{2}$, where the first hypergeometric factor degenerates to 1; the two agree to twelve decimal places.

Remark 10.4. Combining (74), (75) and (77) at Zhuravlev's own $v_p = p|f(p)|$ gives $\mathcal{A}(\mathbf{0}) = 1.1218\dots$, so that his coefficients satisfy

$$S_b(1, U) = \frac{0.8052\dots}{\log U} + o\left(\frac{1}{\log U}\right), \quad \text{whence} \quad \kappa_0 \geq 0.0571\dots$$

These are of course not the numbers Zhuravlev states. His theorem gives $\kappa_0 \geq 0.0248493\dots$ in the corrected form of Corollary 11.6, and the two differ for a reason that has nothing to do with his choice of f : he estimates $S_b(1, U)$ by Wirsing's mean-value theorem, which was the natural tool for a sum of this shape in 1974, whereas the calculation above evaluates the same sum exactly, using the four-variable Euler product of Section 7. Passing from $S_b(1, U)$ to κ_0 also uses Section 3.3 and the coefficient-uniform estimates of Section 5. What the calculation shows is that his coefficients were well chosen. Wirsing's inequality accounts for a factor 2.3 on its own; optimising v_p prime by prime inside (74) moves the constant by under one per cent, and $f(2), f(3), f(5), f(7)$ are already within a few per cent of the minimisers. By the local inequality of Section 11 the multiplicative structure offers nothing further either. The freedom that remains is the profile, and that is what Sections 10 and 11 exploit.

It is natural to ask whether one gains by keeping Zhuravlev's coefficients in place of λ . Ordering the constants of (75), smallest first,

three squared mollifiers (Theorem 11.5)	0.6568	$\kappa_0 > 0.0700162$
$\lambda, a = \frac{1}{4}$	0.6783	$\kappa_0 > 0.0677586$
$\lambda, a = \frac{1}{2}$	0.7178	$\kappa_0 > 0.0640660$
Zhuravlev's f	0.8052	$\kappa_0 > 0.0571$
λ , sharp cutoff	0.8869	$\kappa_0 > 0.0518466$
Wirsing's bound on Zhuravlev's f	1.8506	$\kappa_0 > 0.0248493$

One does not: his choice sits inside the family treated here, with profile exponent $\frac{1}{2}$ in place of the more favourable $\frac{1}{4}$, and with the arithmetic factor $\mathcal{A}(\mathbf{0}) > 1$ of (76) attached to squarefree support, which is independent of the profile and so cannot be traded against it. Relaxing either brings one back to λ at $a = \frac{1}{4}$, and the positive-semidefinite programme improves on that in turn. It is worth noting how close the 1974 choice comes: it already lies above the sharp-cutoff model of Section 6.

10.2 The quarter-power constant

The four-dimensional integral admits the following one-dimensional reduction. Put $\alpha = 1/4$ and $q = a + 1/2$. After setting $y = 1 - \tau_{14}$ and $z = 1 - \tau_{23}$, the integrations in τ_{13} and τ_{24} each produce

$$H_a(y, z) := \int_0^{\min(y, z)} t^{\alpha-1} (y-t)^{q-1} (z-t)^{q-1} dt.$$

If $m = \min(y, z)$ and $M = \max(y, z)$, Euler's beta integral gives

$$H_a(y, z) = B(\alpha, q) m^{\alpha+q-1} M^{q-1} {}_2F_1(1-q, \alpha; \alpha+q; m/M). \quad (78)$$

Using the symmetry in y, z , restricting to $0 < z < y < 1$, and writing $z = yt$, we therefore obtain

$$\begin{aligned} & \int_{\Omega} (\tau_{13}\tau_{14}\tau_{23}\tau_{24})^{-3/4} (c_1c_2c_3c_4)^{a-1/2} d\boldsymbol{\tau} \\ &= 2B\left(\frac{1}{4}, a + \frac{1}{2}\right)^2 B\left(4a + \frac{1}{2}, \frac{1}{4}\right) \int_0^1 t^{2a-1/2} {}_2F_1\left(\frac{1}{2} - a, \frac{1}{4}; a + \frac{3}{4}; t\right)^2 \\ & \quad \times {}_2F_1\left(\frac{3}{4}, 4a + \frac{1}{2}; 4a + \frac{3}{4}; t\right) dt. \end{aligned} \quad (79)$$

Indeed, after (78) the remaining y -integral is

$$\int_0^1 y^{4a-1/2} (1-y)^{-3/4} (1-yt)^{-3/4} dy = B\left(4a + \frac{1}{2}, \frac{1}{4}\right) {}_2F_1\left(\frac{3}{4}, 4a + \frac{1}{2}; 4a + \frac{3}{4}; t\right).$$

For $a = 1/4$, Euler's transformation gives

$${}_2F_1\left(\frac{3}{4}, \frac{3}{2}; \frac{7}{4}; t\right) = (1-t)^{-1/2} {}_2F_1\left(1, \frac{1}{4}; \frac{7}{4}; t\right).$$

Substitution into (79), together with $K_{1/4} = \Gamma(5/4)/\Gamma(3/4)$ and the reflection formula, yields

$$\mathcal{C}_{1/4} = \frac{\pi^{7/2}\sqrt{2}}{96\Gamma(3/4)^6} \int_0^1 (1-t)^{-1/2} F(t)^2 G(t) dt, \quad (80)$$

where

$$F(t) = {}_2F_1\left(\frac{1}{4}, \frac{1}{4}; 1; t\right), \quad G(t) = {}_2F_1\left(1, \frac{1}{4}; \frac{7}{4}; t\right).$$

Write $F(t)^2 G(t) = \sum_{n \geq 0} q_n t^n$; all q_n are non-negative. Then

$$J = \sum_{n \geq 0} q_n B(n+1, 1/2), \quad \sum_{n \geq 0} q_n = \frac{3\pi}{2\Gamma(3/4)^4}.$$

Since the beta factors decrease,

$$\begin{aligned} \sum_{n=0}^N q_n B(n+1, 1/2) &\leq J \leq \sum_{n=0}^N q_n B(n+1, 1/2) \\ &\quad + B(N+2, 1/2) \left(\frac{3\pi}{2\Gamma(3/4)^4} - \sum_{n=0}^N q_n \right). \end{aligned}$$

Exact rational coefficient generation followed by Arb ball arithmetic [12] (python-flint 0.9.0, 256-bit precision) at $N = 700$ gives

$$0.6779054 < \mathcal{C}_{1/4} < 0.6786576.$$

In particular

$$\frac{1}{8e\mathcal{C}_{1/4}} > 0.0677586. \quad (81)$$

10.3 Regularisation at the final coefficient and the order of limits

The power weight vanishes at the final prime, so to use Lemma 5.4 we first regularise it. For fixed $\varepsilon > 0$ put

$$P_{\varepsilon, a}(x) = \left(\frac{x + \varepsilon}{1 + \varepsilon} \right)^a, \quad b_{n, a, \varepsilon}(U) = \lambda(n) P_{\varepsilon, a}\left(\frac{\log(U/n)}{\log U} \right). \quad (82)$$

Then $|b_{n,a,\varepsilon}(U)| \leq 1$ and, for prime U ,

$$|b_{U,a,\varepsilon}(U)| = \frac{1}{2} \left(\frac{\varepsilon}{1+\varepsilon} \right)^a > 0. \quad (83)$$

For fixed ε , this lower bound verifies (33) as $T \rightarrow \infty$.

Let

$$L_{a,\varepsilon,R}(c) = \sum_{n \leq e^{Rc}} \frac{\lambda(n)}{n} \left(\frac{c - \log n/R + \varepsilon}{1+\varepsilon} \right)^a.$$

Stieltjes summation, including the non-zero endpoint term, gives

$$L_{a,\varepsilon,R}(c) = \frac{1}{(1+\varepsilon)^a} \left\{ \varepsilon^a L(e^{Rc}) + a \int_0^c L(e^{R(c-v)})(v+\varepsilon)^{a-1} dv \right\}. \quad (84)$$

Hence, for fixed $\varepsilon > 0$ and locally uniformly for $c > 0$,

$$R^{1/2} L_{a,\varepsilon,R}(c) \longrightarrow \Psi_{a,\varepsilon}(c), \quad (85)$$

where

$$\Psi_{a,\varepsilon}(c) = \frac{1}{\sqrt{\pi}(1+\varepsilon)^a} \left\{ \varepsilon^a c^{-1/2} + a \int_0^c (c-v)^{-1/2} (v+\varepsilon)^{a-1} dv \right\}. \quad (86)$$

Let $S_{a,\varepsilon}(U)$ denote (16) for the coefficients in (82). The same proof as Lemmas 10.1 and 10.2 gives

$$S_{a,\varepsilon}(U) = \frac{\mathcal{C}_{a,\varepsilon}}{\log U} + o_{a,\varepsilon} \left(\frac{1}{\log U} \right), \quad (87)$$

with

$$\mathcal{C}_{a,\varepsilon} = \frac{1}{\Gamma(1/4)^4} \int_{\Omega} \prod_{e \in E} \tau_e^{-3/4} \prod_{i=1}^4 \Psi_{a,\varepsilon}(c_i) d\tau. \quad (88)$$

Moreover,

$$\mathcal{C}_{a,\varepsilon} \longrightarrow \mathcal{C}_a \quad (\varepsilon \downarrow 0). \quad (89)$$

Indeed, (86) converges pointwise to $K_a c^{a-1/2}$, because

$$\frac{a}{\sqrt{\pi}} \int_0^c (c-v)^{-1/2} v^{a-1} dv = \frac{\Gamma(a+1)}{\Gamma(a+1/2)} c^{a-1/2}.$$

For $0 < \varepsilon \leq 1$, (86) is bounded by $C_a(c^{-1/2} + c^{a-1/2})$. The sharp polyhedral integral already proves the integrability of the first term, and the second is weaker. Dominated convergence therefore proves (89).

Proposition 10.5. *Fix $0 < \theta < 1/4$ and $\varepsilon > 0$. For every sufficiently large T choose, by Bertrand's postulate, a prime*

$$T^\theta \leq U_T \leq 2T^\theta.$$

Then $\log U_T = \theta \log T + O(1)$, the coefficients (82) satisfy the hypotheses of Proposition 5.8, and that proposition may be applied with the constant $\mathcal{C}_{a,\varepsilon}$. One first lets $T \rightarrow \infty$ with θ, ε fixed, then lets $\varepsilon \downarrow 0$, and only then lets $\theta \uparrow 1/4$. No estimate uniform in ε is required.

Proof. For fixed ε , (83) gives $\log^+(1/|b_U|) = O_{a,\varepsilon}(1)$, so (33) holds; all remaining coefficient bounds are uniform. Equation (87) supplies the diagonal constant as $T \rightarrow \infty$. After taking the resulting liminf in T , equation (89) permits $\varepsilon \downarrow 0$. The restriction $\theta < 1/4$ is open, so the final lower bound is obtained by $\theta \uparrow 1/4$. $\square$

11 Positive-semidefinite profile optimisation

The scalar weight $|M_U(\frac{1}{2} + it)|^2$ imposes a rank-one condition on the coefficient array entering Zhuravlev's quadratic form. Selberg's sign argument requires only non-negativity, so one may instead take a finite sum of squares. For $1 \leq j \leq J$, let

$$M_j(s) = \sum_{n \leq U} b_n^{(j)} n^{-s}, \quad W(t) = \sum_{j=1}^J |M_j(\frac{1}{2} + it)|^2.$$

Put $\mathbf{b}_n = (b_n^{(1)}, \dots, b_n^{(J)})$ and

$$C_{\ell m} = \langle \mathbf{b}_\ell, \mathbf{b}_m \rangle. \quad (90)$$

Then C is real symmetric and positive semidefinite, and

$$W(t) = \sum_{\ell, m \leq U} \frac{C_{\ell m}}{\sqrt{\ell m}} \left(\frac{m}{\ell}\right)^{it} \geq 0. \quad (91)$$

Conversely every real positive-semidefinite matrix has a Gram factorisation. Define

$$Y_C(s) = \zeta(s) \sum_{\ell, m \leq U} C_{\ell m} \ell^{-s} m^{s-1}. \quad (92)$$

Symmetry gives $Y_C(s) = \chi(s) Y_C(1-s)$, while on the critical line $\chi(s)^{-1/2} Y_C(s) = Z(t) W(t)$. Thus the new detector preserves the sign of Hardy's function. We impose

$$C \succeq 0, \quad C_{11} = 1, \quad C_{nn} \leq 1 \quad (n \leq U). \quad (93)$$

The Gram-matrix Cauchy–Schwarz inequality then gives $|C_{\ell m}| \leq 1$.

Proposition 11.1. *Suppose that $C = C(U)$ satisfies (93) and that*

$$C_{1U} \neq 0, \quad \log^+ \frac{1}{|C_{1U}|} \ll \log T. \quad (94)$$

Then the coefficient-uniform analytic proof of Proposition 4.3 applies to Y_C , with the same factor $c = 2$ and the same range $U = T^\theta$, $\theta < 1/4$. If $S_C(1, U) \leq (\gamma + o(1))/\log U$, then $\kappa_0 \geq 2\theta/(ec^2\gamma)$.

Proof. The regularised approximate functional equation is linear in the array multiplying $\ell^{-s} m^{s-1}$, and the fixed-phase construction extends from $b_\ell b_m$ to $C_{\ell m}$. Choose Gram coordinates with $\mathbf{b}_1 = (1, 0, \dots, 0)$; then $W(t) \geq |M_1(\frac{1}{2} + it)|^2$, giving the same critical-edge lower bound. The large-disk estimate uses only $|C_{\ell m}| \leq 1$.

For the centre estimate the unique smallest frequency $1/U$ comes from $(n, \ell, m) = (1, 1, U)$ and has coefficient C_{1U}/U . Repeating Appendix A with b_U replaced by C_{1U} gives, for the same centre s_0 ,

$$\Lambda_C^+(s_0) = C_{1U} U^{s_0-1} \{1 + O(e^{-cL^2})\},$$

where (94) is exactly what is needed to dominate the remaining frequencies and the polar term. The mean-square expansion is quadratic in C ; its diagonal is Proposition 11.2. The secondary form is

$$Q_C(\alpha, U) = \sum_{d \leq U} \Phi_\alpha(d) \left(\sum_{\substack{\ell, m \leq U \\ d|\ell m}} \frac{C_{\ell m}}{\ell^\alpha m} \right)^2 \geq 0,$$

so the coefficient $\zeta(\alpha)$ again has the favourable sign. The rational-frequency off-diagonal estimate is unchanged, and the logarithmic optimisation is identical to the scalar case. $\square$

Proposition 11.2. *For a symmetric coefficient matrix C ,*

$$S_C(1, U) = \sum_{d \leq U} \frac{\phi(d)}{d^2} \left(\sum_{\substack{\ell, m \leq U \\ d | \ell m}} \frac{C_{\ell m}}{\ell m} \right)^2. \quad (95)$$

For fixed U , this is a convex quadratic function of C .

Proof. The scalar diagonal is obtained by grouping equal rational frequencies and using $\sum_{d | (n_1, n_2)} \phi(d) = (n_1, n_2)$. The calculation is linear in each occurrence of $b_\ell b_m$, so replacing it by $C_{\ell m}$ gives (95). Each inner sum is linear in C , and S_C is a sum of squares. $\square$

The unrestricted finite problem

$$\min\{S_C(1, U) : C \succeq 0, C_{11} = 1, C_{nn} \leq 1\}$$

is convex, but its asymptotic arithmetic structure is not transparent. We retain the reciprocal-square-root sequence and enlarge only the logarithmic profile. Put

$$v_n = \frac{\log(U/n)}{\log U}, \quad \mathbf{b}_n = \lambda(n)q(v_n),$$

where $q : [0, 1] \rightarrow \mathbb{R}^J$, $\|q(1)\| = 1$, and $\|q(v)\| \leq 1$. The associated kernel is $K(u, v) = \langle q(u), q(v) \rangle$.

Let $\mathcal{P}$ be the class of scalar functions

$$f(v) = f_0 + \sum_{r=1}^m f_r v^{a_r}, \quad a_r > 0,$$

and let $\mathcal{P}^J$ denote vector-valued profiles whose components lie in $\mathcal{P}$. For $f \in \mathcal{P}$ define the endpoint-inclusive transform

$$\tilde{\Psi}_f(c) = \frac{1}{\sqrt{\pi}} \left\{ f(0)c^{-1/2} + \int_0^c (c-v)^{-1/2} f'(v) dv \right\}. \quad (96)$$

Equivalently,

$$\tilde{\Psi}_f(c) = \frac{f_0}{\sqrt{\pi c}} + \sum_{r=1}^m f_r \frac{\Gamma(a_r + 1)}{\Gamma(a_r + 1/2)} c^{a_r - 1/2}. \quad (97)$$

For $f_1, f_2, f_3, f_4 \in \mathcal{P}$, define

$$S_{f_1, f_2; f_3, f_4}(U) = \sum_{d \leq U} \frac{\phi(d)}{d^2} \left(\sum_{\substack{\ell, m \leq U \\ d | \ell m}} \frac{\lambda(\ell)\lambda(m)f_1(v_\ell)f_2(v_m)}{\ell m} \right) \times \left(\sum_{\substack{n, r \leq U \\ d | nr}} \frac{\lambda(n)\lambda(r)f_3(v_n)f_4(v_r)}{nr} \right). \quad (98)$$

Mixed four-profile transfer. For fixed $f_1, f_2, f_3, f_4 \in \mathcal{P}$,

$$S_{f_1, f_2; f_3, f_4}(U) = \frac{\mathcal{C}(f_1, f_2, f_3, f_4) + o(1)}{\log U},$$

where

$$\mathcal{C}(f_1, f_2, f_3, f_4) = \frac{1}{\Gamma(1/4)^4} \int_{\Omega} \prod_{e \in E} \tau_e^{-3/4} \prod_{i=1}^4 \tilde{\Psi}_{f_i}(c_i) d\tau. \quad (99)$$

The assertion is multilinear and the error is uniform when the finitely many coefficients and positive exponents range over a fixed compact set.

Proof. Write $R = \log U$ and

$$W_{f,R}(c) = R^{1/2} \sum_{n \leq e^{Rc}} \frac{\lambda(n)}{n} f\left(c - \frac{\log n}{R}\right).$$

Stieltjes summation gives the exact identity

$$W_{f,R}(c) = R^{1/2} f(0) L(e^{Rc}) + R^{1/2} \int_0^c L(e^{R(c-v)}) f'(v) dv.$$

the scalar Mertens-sum and power estimates of Sections 9 and 10 imply local uniform convergence to $\tilde{\Psi}_f(c)$. By (97), the absolute value is bounded by a fixed finite sum of terms $(c + R^{-1})^{-1/2}$ and $(c + R^{-1})^{a-1/2}$. The polyhedral majorant argument of Section 9 therefore gives a common integrable boundary majorant, including every face on which one or more deficits c_i vanish.

Expanding the four positive cross factors of the polar model produces the positive edge measures of the residue calculation and the four signed tails $W_{f_i,R}(c_i)$. Interior convergence and the preceding majorant give (99) for the polar model, uniformly when the coordinate caps lie near 1. Finally restore the regular Euler factor A . Its coefficients satisfy weighted ℓ^1 summability, the tail $\max r_i > e^{\sqrt{R}}$ is $o(R^{-1})$, and on the complementary range the uniform polar limit and continuity at $\mathbf{b} = \mathbf{1}$ apply, with $\sum a(\mathbf{r}) = A(0) = 1$. This proves the claim. $\square$

Theorem 11.3. *Let $q \in \mathcal{P}^J$, with $\|q(1)\| = 1$ and $\|q(v)\| \leq 1$. Define $\tilde{\Psi}_q$ componentwise by (96). Then*

$$S_q(1, U) = \frac{C[q] + o(1)}{\log U},$$

where

$$C[q] = \frac{1}{\Gamma(1/4)^4} \int_{\Omega} \prod_{e \in E} \tau_e^{-3/4} \langle \tilde{\Psi}_q(c_1), \tilde{\Psi}_q(c_2) \rangle \langle \tilde{\Psi}_q(c_3), \tilde{\Psi}_q(c_4) \rangle d\tau. \quad (100)$$

The functional depends only on the positive-semidefinite kernel $K(u, v) = \langle q(u), q(v) \rangle$ and is convex quadratic in that kernel.

Proof. Write $q = (q_1, \dots, q_J)$. Expanding (95) gives a sum over r, s of products of two inner sums in q_r and q_s . Apply the mixed four-profile transfer above to (q_r, q_r, q_s, q_s) and sum over r, s ; this yields the two inner products in (100). For finite U , Proposition 11.2 is a sum of squares of linear functionals of K , and passage to the limit preserves convexity. $\square$

Remark 11.4. The power profiles below satisfy $q(0) = 0$, so their terminal coefficient vanishes. For fixed $\varepsilon > 0$ put $q_\varepsilon(v) = (q(v) + \varepsilon q(1))/(1 + \varepsilon)$. Then $q_\varepsilon(1) = q(1)$, convexity of the unit ball gives $\|q_\varepsilon(v)\| \leq 1$, and, when U is prime, $C_{1U} = \langle q_\varepsilon(1), \lambda(U) q_\varepsilon(0) \rangle = \lambda(U) \varepsilon / (1 + \varepsilon) \neq 0$, so Proposition 11.1 applies. The corrected transform is

$$\tilde{\Psi}_{q_\varepsilon}(c) = \frac{1}{1 + \varepsilon} \left\{ \Psi_q(c) + \frac{\varepsilon q(1)}{\sqrt{\pi c}} \right\}, \quad (101)$$

where Ψ_q is the derivative integral of the unregularised profile. The additional $c^{-1/2}$ term is controlled by the sharp-profile majorant, so dominated convergence in (100) gives $C[q_\varepsilon] \rightarrow C[q]$ as $\varepsilon \downarrow 0$; no uniformity in ε is required before the $T \rightarrow \infty$ limit.

For power directions $p_i(v) = v^{a_i}$, let $p(v) = (p_1(v), \dots, p_m(v))^T$ and take $M \succeq 0$ with

$$\mathbf{1}^T M \mathbf{1} = 1, \quad p(v)^T M p(v) \leq 1 \quad (0 \leq v \leq 1). \quad (102)$$

Then $K_M(u, v) = p(u)^T M p(v)$ and

$$C(M) = \sum_{i,j,k,l=1}^m M_{ij} M_{kl} T(a_i, a_j, a_k, a_l), \quad (103)$$

where

$$T(a_1, a_2, a_3, a_4) = \frac{\prod_{r=1}^4 \Gamma(a_r + 1) / \Gamma(a_r + 1/2)}{\Gamma(1/4)^4} \int_{\Omega} \prod_{e \in E} \tau_e^{-3/4} \prod_{r=1}^4 c_r^{a_r - 1/2} d\tau. \quad (104)$$

Thus a fixed power basis gives a convex quadratic semidefinite programme. The tensor reduction and rigorous evaluation are recorded in Appendix B.

Theorem 11.5. *There is an explicit nine-direction profile $q_3 : [0, 1] \rightarrow \mathbb{R}^3$, given in Appendix B, with $\|q_3(v)\| \leq 1$ on $[0, 1]$ and*

$$C[q_3] = 0.6567752140190419405677628751 \dots$$

with rigorous interval radius below 10^{-17} . The corresponding sum of three squared mollifiers satisfies $\kappa_0 > 0.0700162$.

Proof. Appendix B gives the exact profile, its Bernstein admissibility certificate and the interval enclosure of $C[q_3]$. For fixed $\varepsilon > 0$, Proposition 11.1 and Theorem 11.3 give the diagonal asymptotic with $C[q_\varepsilon]$. Letting $T \rightarrow \infty$, then $\varepsilon \downarrow 0$ via Remark 11.4, and finally $\theta \uparrow 1/4$, gives $\kappa_0 \geq 1/(8eC[q_3]) > 0.0700162$. $\square$

The improvement is a profile-kernel effect rather than a local Euler-factor correction. If $C(z) = \sum_{\nu \geq 0} c_\nu z^\nu$ with $C(0) = 1$ and $C(z)^2 = \sum_{r \geq 0} a_r z^r$, then with $A_r = \sum_{k \leq r} a_k$ the local diagonal factor is $(1 - p^{-1}) \sum_{r \geq 0} p^{-r} A_r^2 \geq 1 - p^{-1}$, with equality precisely when $C(z)^2 = 1 - z$, that is, for the local coefficients of $\zeta(s)^{-1/2}$. Reciprocal-square-root arithmetic is therefore already prime-by-prime optimal within the multiplicative scalar class; the gain above comes from correlating several logarithmic profiles.

This inequality is sharper than it looks, and it fixes what the improvements in this paper should be credited to. In the notation of Section 10.1 the local factor is $\mathcal{A}_{b,p}(\mathbf{0}) = \sum_{r \geq 0} A_r^2 p^{-r}$, so the arithmetic factor $\mathcal{A}_b(\mathbf{0})$ of (75) is at least 1 for every multiplicative b , with equality if and only if $b * b = \mu$. The convolution identity (37), which entered as a device for removing the same-side poles, is thus exactly the condition for the arithmetic factor to be optimal, and λ is its unique solution. Consequently the whole of the remaining freedom lies in $\mathcal{C}_a$, and the improvements recorded below are profile effects and nothing else.

The comparison with Zhuravlev is best read in that light. His constant $1.8505\dots$ is Wirsing's upper bound for the diagonal; the diagonal itself, at his coefficients, is $0.8052\dots$ by Remark 10.4. Of the factor 2.818 separating Corollary 11.6 from Theorem 11.5, a factor 2.298 is thus the passage from that bound to an exact evaluation and a factor 1.226 is the better profile. The sharp coefficients of Section 6, taken with a sharp cutoff, give $0.8869\dots$, so they sit about ten per cent below Zhuravlev's choice, whose normalisation carries the $a = \frac{1}{2}$ profile without comment; the arithmetic here overtakes his only at $a = \frac{1}{4}$ and beyond.

11.1 Critical-line proportions

These are the proportions the transfer theorems above deliver; they are what Section 1.2 quotes.

Corollary 11.6. *Assume the coefficient estimate (8) from Zhuravlev's Wirsing calculation. With the source normalisation $\aleph(s)Y_U(s) = 2 \operatorname{Re} \beta_U(s)$ and $\theta \uparrow \frac{1}{4}$, his method proves*

$$\kappa_0 \geq \frac{2}{3\pi^2 e} = 0.0248493202 \dots$$

The printed bound $2/21$ results from omitting the term $-T \log 2$ and is not a consequence of the stated definitions.

Proof. The coefficient bounds preceding Proposition 3.1 give $|b_n| \leq 1$, while (7) gives $\log^+(1/|b_U|) \ll \log U$. Hence (33) holds for $U \asymp T^\theta$, $\theta < 1/4$, and Proposition 5.8 applies with $\gamma = 3\pi^2/16$. Proposition 4.3, with $c = 2$, therefore gives

$$\kappa_0 \geq \frac{2\theta}{e 4(3\pi^2/16)}.$$

Letting $\theta \uparrow 1/4$ yields $2/(3\pi^2 e)$. □

Theorem 11.7. *Use Zhuravlev's original normalisation*

$$\aleph(s)Y_U(s) = 2 \operatorname{Re} \beta_U(s) \quad (\operatorname{Re} s = \tfrac{1}{2}).$$

(i) *For the sharp coefficients $b_n = \lambda(n)$,*

$$\kappa_0 \geq \frac{1}{8e\mathcal{C}_{\text{res}}} = \frac{\Gamma(\frac{3}{4})^4}{16e} = 0.0518466520\dots$$

(ii) *For the regularised quarter-power coefficients and then $\varepsilon \downarrow 0$,*

$$\kappa_0 \geq \frac{1}{8e\mathcal{C}_{1/4}} > 0.0677586.$$

(iii) *For the three-square positive-semidefinite profile of Theorem 11.5,*

$$\kappa_0 > 0.0700162.$$

Proof. Fix $0 < \theta < \frac{1}{4}$. For the sharp coefficients, choose a prime $U \asymp T^\theta$. The coefficient hypotheses in Proposition 5.8 hold because $|\lambda(n)| \leq 1$ and $\lambda(U) = -\frac{1}{2}$. Theorem 9.6 gives

$$S_\lambda(U) = \frac{\mathcal{C}_{\text{res}} + o(1)}{\log U}.$$

Propositions 5.8 and 4.3, with $c = 2$, therefore yield

$$\kappa_0 \geq \frac{\theta}{2e\mathcal{C}_{\text{res}}}.$$

Letting $\theta \uparrow \frac{1}{4}$ proves part (i). The exact form follows from Theorem 8.1.

For the power weight, fix $\varepsilon > 0$ and use the prime selection in Proposition 10.5. Equations (87) and (89) give, successively,

$$\kappa_0 \geq \frac{\theta}{2e\mathcal{C}_{a,\varepsilon}}, \quad \kappa_0 \geq \frac{\theta}{2e\mathcal{C}_a}.$$

Letting $\theta \uparrow \frac{1}{4}$, taking $a = \frac{1}{4}$, and using the certified upper bound for $\mathcal{C}_{1/4}$ proves part (ii). Part (iii) is Theorem 11.5. □

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Declaration on the use of AI

The English translation of Zhuravlev's 1974 paper was produced with the assistance of a large language model. The positive-semidefinite optimisation of Section 11, and the interval-arithmetic verification of its constant in Appendix B, were carried out with the same assistance. The author has checked these outputs and is responsible for the content of the paper.

Appendix A Uniform bounds for Lavrik's analytic half

This appendix supplies the large-disk estimates used in Lemma 5.4. They are needed, and cannot be quoted from the literature, for the following reason. The Jensen step of Section 4 is applied on a disk whose radius grows with T , and it therefore requires a bound on $\Lambda_U(s)$ that is uniform over that whole disk and uniform in the mollifier coefficients b_n . The approximate functional equations available in the literature are stated in a fixed vertical strip, and extrapolating one of them to a growing disk is exactly the step that is not justified. The point of using the explicit regularised incomplete-gamma kernel is that every singularity and every dependence on the terminal coefficient is visible; no fixed-strip approximate functional equation is extrapolated to the Jensen disk.

A.1 The Mellin contour and its singularities

Use the principal logarithm in $X^{-z} = \exp(-z \log X)$, where $|\arg X| < \pi/4$, and begin in (19) on the vertical line $\operatorname{Re} z = \Delta > \max(0, \operatorname{Re} s)$. For $X > 0$ and $\operatorname{Re} s > 0$, differentiation under the integral and Mellin inversion give

$$\frac{\partial}{\partial X} J(s, X) = -2X^{s-1} e^{-X^2}.$$

On the initial line $J(s, X) = O(X^{-\Delta})$ as $X \rightarrow +\infty$. Integration from X to infinity therefore gives

$$J(s, X) = \Gamma(s/2, X^2).$$

Analytic continuation in X through $|\arg X| < \pi/4$, and then in s , proves the identity in (20) without a limiting shift to a receding left vertical line.

For completeness, a finite shift of the Mellin line records the singularities used below. For generic s , shifting past N gamma poles crosses exactly

$$z = 0, \quad z = -s - 2k \quad (0 \leq k < N).$$

Their respective residues are

$$\Gamma(s/2), \quad -\frac{2(-1)^k X^{s+2k}}{k!(s+2k)}.$$

Stirling's formula justifies each finite shift and makes its two closing horizontal pieces tend to zero. The quotient $F(s, X) = \Gamma(s/2, X^2)/\Gamma(s/2)$ is entire in s . The only singularity of $\Lambda_U^\pm(s)$ in (21) is the displayed simple pole at $s = 1$; consequently the only singularities of $G_\pm(s)$ are at 1 and $1 + 2it$. This proves, in particular, that

$$H_\pm(s) = (s-1)(s-1-2it)G_\pm(s) \tag{A.1}$$

is entire. The residue calculation also records all poles crossed in passing from the Mellin contour to the kernel representation used below.

A.2 Uniform kernel estimates

Put $\phi_T = \pi/2 - 1/T$, and write $\rho_T(v) = \rho_T^+$ for $v > 0$ and $\rho_T(v) = \rho_T^-$ for $v < 0$. If $s = \sigma + iv$, $|v| \asymp T$, and $|\sigma| = o(T^{1/2})$, integration along the ray $u = re^{\operatorname{sgn}(v)i\phi_T}$, followed by uniform Stirling, gives

$$|F(s, (\pi\rho_T(v))^{1/2}\xi)| \leq C^{|\sigma|+1} T^{1/2} \int_{c\xi^2/T}^\infty e^{-u} u^{\sigma/2-1} du. \tag{105}$$

Here and below $c, C > 0$ are absolute. Indeed, the numerator contributes $e^{-|v|\phi_T/2}$, while

$$|\Gamma((\sigma + iv)/2)|^{-1} \ll T^{1/2-\sigma/2} e^{\pi|v|/4};$$

because $\pi/2 - \phi_T = 1/T$, the two exponential factors cancel up to an absolute constant. This is the reason that the phases $\rho_T^\pm$ were fixed as in Section 5.

For $\sigma > 0$, the complementary lower-gamma integral gives, uniformly for $0 < \xi \leq 1$,

$$F(s, (\pi\rho_T(v))^{1/2}\xi) = 1 + O\left\{C^\sigma T^{1/2} \left(\frac{\xi^2}{T}\right)^{\sigma/2}\right\}. \quad (106)$$

The estimates remain uniform if v varies by $o(T)$. They are direct estimates of the old contour after the complete residue calculation above; there are no unrecorded vertical or horizontal contour pieces.

We shall also use two elementary consequences of (18). First, for $\sigma \geq 2$,

$$\sum_{\xi > 1} |a_\xi| \xi^{-\sigma} \ll U \log(2U). \quad (107)$$

To see this, fix ℓ, m . If $m \geq \ell$, integral comparison in $n > m/\ell$ gives a contribution $\ll 1/\ell + 1/m$; if $m < \ell$, the same conclusion follows directly from $(m/\ell)^\sigma \zeta(\sigma)/m$. Summation over $\ell, m \leq U$ proves (107). Second, we prove explicitly the frequency sum needed on the Jensen disk. Whenever $|\sigma| = o(T^{1/2})$,

$$\sum_{\xi > 0} |a_\xi| \xi^{-\sigma} |F(s, (\pi\rho_T(v))^{1/2}\xi)| \leq \exp\{C(|\sigma| + U) \log(2T)\}. \quad (108)$$

Indeed, put $q = |\sigma|$ and $y = c\xi^2/T$. Every frequency in (18) satisfies $\xi \geq 1/U$, so $y \gg (TZ^2)^{-1}$. Splitting the incomplete-gamma integral at 1 and at $q+2$ gives the elementary uniform estimates

$$\begin{aligned} \int_y^\infty e^{-u} u^{\sigma/2-1} du &\ll L \exp\{Cq \log(q+2)\} e^{-cy}, & \sigma \geq 0, \\ \int_y^\infty e^{-u} u^{\sigma/2-1} du &\ll Ly^{-q/2} e^{-cy}, & \sigma < 0, \end{aligned}$$

where $L = \log(2T)$. In the second line the factor $y^{-q/2}$ cancels ξ^q after (105); in the first line we use $\xi^{-q} \leq U^q$. Since $q = o(T^{1/2})$, in either case

$$\xi^{-\sigma} |F(s, (\pi\rho_T(v))^{1/2}\xi)| \leq \exp\{C(q+1)L\} e^{-c\xi^2/T}. \quad (109)$$

Finally, expanding a_ξ and using $\sum_{n \geq 1} e^{-An^2} \ll 1 + A^{-1/2}$,

$$\begin{aligned} \sum_{\xi > 0} |a_\xi| e^{-c\xi^2/T} &\leq \sum_{\ell, m \leq U} \frac{1}{m} \sum_{n \geq 1} e^{-c(n\ell/m)^2/T} \\ &\ll \sum_{\ell, m \leq U} \left(\frac{1}{m} + \frac{\sqrt{T}}{\ell} \right) \ll U\sqrt{T} \log(2U). \end{aligned}$$

Combining this with (109) proves (108), with room to spare.

Proposition A.1. *Let the hypotheses of Lemma 5.4 hold and put*

$$\sigma_0 = AU\{L^2 + \eta_U + 1\} + \sigma_1 + \sigma_2, \quad s_0 = \sigma_0 + it,$$

where A is a sufficiently large absolute constant. Then

$$\max_{|s-s_0| \leq 2\sigma_0} \log^+ |H_\pm(s)| \ll ZL\{L^2 + \eta_U + 1\}. \quad (110)$$

For at least one choice of sign,

$$-\log |H_\pm(s_0)| \ll ZL\{L^2 + \eta_U + 1\}. \quad (111)$$

The constants are uniform for $T \leq t \leq 2T$.

Proof. Since $\eta_U \ll L$, we have $\sigma_0 \ll ZL^2 = o(T^{1/2})$. Throughout the disk in (110), the imaginary parts of s and $s - 2it$ have absolute value $\asymp T$. Apply (108) to the two halves. The polar term in (21) is bounded there by the same right side, since $|M_U(0)M_U(1)| \ll U \log(2U)$. The two regularising linear factors cost only $O(L + \log \sigma_0)$ on the logarithmic scale. As $|\operatorname{Re} s| \leq 3\sigma_0$, this proves (110).

It remains to prove the centre estimate. The smallest frequency in (18) is $1/U$, it occurs only for $(n, \ell, m) = (1, 1, U)$, and its coefficient is b_U/U . Every other frequency at most one is at least $1/(U-1)$. From (106), (107), and Stirling,

$$\begin{aligned} \Lambda_U^+(s_0) &= b_Z Z^{s_0-1} + E_1 + E_2 + E_3, |E_1| \ll U^3(U-1)^{\sigma_0}, \\ |E_2| &\ll U \log(2U) C^{\sigma_0} T^{1/2} \Gamma(\sigma_0/2), \\ |E_3| &\ll U \log(2U) C^{\sigma_0} T^{-1/2-\sigma_0/2}. \end{aligned}$$

Here E_1 contains the other frequencies at most one (and the uniform error in replacing their kernels by one), E_2 contains $\xi > 1$, and E_3 is the polar term. The deliberately coarse factor U^3 in E_1 dominates the total absolute coefficient mass in $0 < \xi \leq 1$.

Now $|b_U| \geq e^{-\eta_U}$ and

$$\log \frac{U}{U-1} \geq \frac{1}{U}.$$

Consequently, after division by $|b_U|U^{\sigma_0-1}$, the logarithm of the first error is at most

$$O(L + \eta_U) - \sigma_0 \log \frac{U}{U-1} \leq O(L + \eta_U) - A\{L^2 + \eta_U + 1\}.$$

For the second error, Stirling's upper bound for $\Gamma(\sigma_0/2)$, together with $\log \sigma_0 = \log U + O(\log L)$, gives

$$\log \frac{|E_2|}{|b_U|U^{\sigma_0-1}} \leq -\frac{\sigma_0}{2} \log U + O(\sigma_0 \log L + \sigma_0 + L + \eta_U),$$

which tends to $-\infty$; the third error is smaller still. Taking A large enough in the first comparison therefore yields

$$\Lambda_U^+(s_0) = b_Z Z^{s_0-1} \{1 + O(e^{-cL^2})\}. \quad (112)$$

The conjugation identity gives $\Lambda_U^-(s_0 - 2it) = \overline{\Lambda_U^+(s_0)}$. Hence at least one of $G_+(s_0), G_-(s_0)$ has modulus at least $|\Lambda_U^+(s_0)|/\sqrt{2}$. Multiplication by the two factors in (A.1), followed by (112), proves (111). $\square$

Appendix B The positive-semidefinite calculation

This appendix gives the finite-basis reduction and the rigorous certificates used in Theorem 11.5. It is self-contained: the coefficient matrix, the closed-form tensor (113), and the Bernstein and contraction data below determine the constant.

B.1 A one-dimensional tensor formula

For $A = a_1 + a_2 + a_3 + a_4$, put

$$K_a = \frac{\Gamma(a+1)}{\Gamma(a+1/2)}, \quad F_{a|b}(t) = {}_2F_1\left(\frac{1}{2} - a, \frac{1}{4}; b + \frac{3}{4}; t\right), \quad H_A(t) = {}_2F_1\left(A, \frac{1}{4}; A + \frac{3}{4}; t\right).$$

Proposition B.1. *The tensor in (104) equals*

$$T(a_1, a_2, a_3, a_4) = \frac{\prod_{r=1}^4 K_{a_r}}{\Gamma(1/4)^4} B\left(A + \frac{1}{2}, \frac{1}{4}\right) (I_1 + I_2), \quad (113)$$

where

$$I_1 = B\left(\frac{1}{4}, a_3 + \frac{1}{2}\right) B\left(\frac{1}{4}, a_2 + \frac{1}{2}\right) \int_0^1 t^{a_2+a_3-1/2} (1-t)^{-1/2} F_{a_1|a_3}(t) F_{a_4|a_2}(t) H_A(t) dt,$$

$$I_2 = B\left(\frac{1}{4}, a_1 + \frac{1}{2}\right) B\left(\frac{1}{4}, a_4 + \frac{1}{2}\right) \int_0^1 t^{a_1+a_4-1/2} (1-t)^{-1/2} F_{a_3|a_1}(t) F_{a_2|a_4}(t) H_A(t) dt.$$

Proof. Integrate two opposite edge variables in Ω by Euler's beta integral, then split according to the ordering of the two residual variables. Scaling the smaller residual variable by the larger produces the two functions $F_{a|b}$; the radial integral is a beta integral, and the final angular integral gives H_A . The two orderings yield I_1 and I_2 . Absolute convergence follows because every exponent in (104) exceeds -1 . $\square$

B.2 The rank-three profile

Take

$$D = (1, 2, 4, 8, 16, 32, 64, 128, 256), \quad a_i = D_i/50 \quad (0 \leq i \leq 8),$$

and write the matrix as the *horizontal concatenation*

$$L_3 = (B_1 \mid B_2 \mid B_3) \in \mathbb{R}^{3 \times 9},$$

where

$$B_1 = \begin{pmatrix} 1.715486756648026 & -2.339134822514005 & -1.111200248286402 \\ -2.139923409444840 & 5.054096020385597 & -4.594675801135149 \\ 0.013952449847396 & 0.030032118739681 & -0.079199051930471 \end{pmatrix}, \quad B_2 = \begin{pmatrix} 4.065763859307927 & -4.609159968314716 & 3.792345042202834 \\ 2.331093987386471 & -0.296257120291222 & -0.133606298590404 \\ -0.113514772779195 & -0.314330575910004 & -0.164466369735625 \end{pmatrix}, \quad B_3 = \begin{pmatrix} -1.496971267659852 & -0.288090695413796 & 0.354925827320853 \\ 0.154467402859762 & 0.184476891646641 & -0.113337159325580 \\ 0.456868764340113 & 0.039242123557055 & -0.038995115885160 \end{pmatrix}.$$

Thus the first row of L_3 is the first row of B_1 , followed by the first rows of B_2 and B_3 ; the blocks are not separately vectorised. Every displayed decimal is interpreted as an exact rational. Put

$$u(x) = (x, x^2, x^4, x^8, x^{16}, x^{32}, x^{64}, x^{128}, x^{256})^T,$$

$$S_3 = \|L_3 \mathbf{1}_9\|_2^2, \quad q_3(v) = S_3^{-1/2} L_3 u(v^{1/50}), \quad M_3 = \frac{L_3^T L_3}{S_3}.$$

The row-sum vector and normalisation are

$$L_3 \mathbf{1}_9 = (0.083964483290869, 0.446334513491276, -0.170410429756210)^T,$$

$$S_3 = 0.235304246957492839754620447437, \quad \mathbf{1}_9^T M_3 \mathbf{1}_9 = 1.$$

In particular $\|q_3(1)\|_2 = 1$ exactly. If a 9×3 factor is desired, it is L_3^T ; it is not obtained by flattening the three blocks.

Proposition B.2. *The profile q_3 satisfies $\|q_3(v)\|_2 \leq 1$ for $0 \leq v \leq 1$.*

Proof. Set $P_3(x) = S_3 - \|L_3 u(x)\|_2^2$. This is an exact rational polynomial of degree 512, with $P_3(1) = 0$. In its degree-512 Bernstein expansion all coefficients are non-negative; the unique zero is the endpoint coefficient b_{512} , and

$$\min_{0 \leq k \leq 511} \frac{b_k}{S_3} = 0.0342355123365906602074626397574 \dots$$

Bernstein positivity gives the claim. $\square$

Proposition B.3. *The interval enclosure of (103) for L_3 satisfies*

$$C[q_3] \in 0.6567752140190419405677628751089899133 \pm 1.0 \times 10^{-17},$$

and therefore

$$\frac{1}{8eC[q_3]} \in 0.0700162386839053965024375047958122649 \pm 1.1 \times 10^{-18}.$$

Proof. For the nine exponent directions, the symmetries $i \leftrightarrow j$, $k \leftrightarrow l$, and $(i, j) \leftrightarrow (k, l)$ leave 1035 tensor orbits. Contracting the exact rational matrix M_3 against the tensor (113), evaluated to 40 digits at these orbits, gives

$$C[q_3] = 0.65677521401904194135363065873002498083982947\dots,$$

which differs from the interval centre by 7.86×10^{-19} . The formal enclosure is the output of the validated interval integration; the arithmetic check verifies the matrix orientation, exact normalisation, Bernstein certificate and contraction centre. $\square$

B.3 The stronger rank-six benchmark

A ten-direction, six-square profile gives the certified enclosure

$$C[q_6] \in 0.6566338678379319741683641732 \pm 5.63 \times 10^{-18},$$

and hence $\frac{1}{8eC[q_6]} > 0.0700313$. The ten-direction tensor has 1540 symmetry orbits; together with the 1035 rank-three orbits this explains the combined total 2575, which should not be attributed to the rank-three contraction alone. We use the rank-three profile in the main theorem because it is substantially simpler and already crosses the seven-per-cent threshold.

B.4 Reproducibility

The constant is reproducible from the data printed above, with no external input. Reading the matrix L_3 as an exact rational, one forms $P_3(x) = S_3 - \|L_3 u(x)\|^2$ and expands it in the degree-512 Bernstein basis to confirm admissibility; one evaluates the tensor (113) at the nine directions $a_i = D_i/50$ and contracts $M_3 = L_3^\top L_3 / S_3$ against it to recover $C[q_3]$. Both steps use only the closed forms of this appendix. The $\pm 10^{-17}$ interval enclosure of Proposition B.3 is the output of a validated interval integration of (113); its complete inputs, logs and software versions may be deposited separately for a permanent record.

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